

Paying by Donating: Corporate Donations Affiliated with Independent Directors

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Corporate donations to charities affiliated with the board's independent directors (*affiliated donations*) are large and mostly undetected due to lack of formal disclosure. Affiliated donations may impair independent directors' monitoring incentives. CEO compensation is on average 9.4% higher at firms making affiliated donations than at other firms, and it is much higher when the compensation committee chair or a large fraction of compensation committee members are involved. We find suggestive evidence that CEOs are unlikely to be replaced for poor performance when firms donate to charities affiliated with a large fraction of the board or when they donate large amounts. (*JEL* G34, J33, M52)

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The monitoring role of independent directors on corporate boards has long been a topic of interest in the corporate governance literature (Weisbach 1988; Hermalin and Weisbach 1998; Core, Holthausen, and Larcker 1999). Stock-exchange rules establishing directors' independence are typically based on financial ties, and most empirical research classifies independent directors based on this limited assessment.¹ Using this conventional definition, the boards of large U.S. companies have been overwhelmingly independent following the majority independence rules enacted by major exchanges in 2002. However, independent directors may have other ties to top executives that interfere with their exercise of independent judgment in carrying out director responsibilities.² This paper explores a new determinant of director independence: material relationships between independent directors and top executives via corporate charitable contributions to tax-exempt organizations affiliated with independent directors (*affiliated donations*).³

In direct contrast to guidelines that govern business transactions with director-affiliated organizations, stock-exchange rules do not bar directors from being classified as independent based on corporate charitable contributions to director-affiliated charities, regardless of their amount. When affiliated donations are made, they are typically larger than the directors' annual compensation. Corporate donations help fulfill directors' fundraising obligations at their affiliated charities, creating a conflict of interest that increases directors' disutility in carrying out monitoring responsibilities (Hermalin and Weisbach 1998; Francis and Lublin 2016).

Because corporate charitable contributions are rarely disclosed in companies' filings with the Securities and Exchange Commission (SEC),⁴ they have been

¹ According to NYSE rules (http://nysemanual.nyse.com/lcm/Help/mapContent.asp?sec=lcm-sections&title=sx-ruling-nyse-policymanual_303A.02&id=chp_1_4_3_3), a director is *not independent* if he or she is a current employee or if the director's immediate family member is a current executive officer of an organization that "receives payments from the company for property or services exceeding the greater of \$1 million, or 2% of the organization's consolidated gross revenue in any of the past three years."

² Recent research examines how independent directors' biographical connections with the CEO (Hwang and Kim 2009; Fracassi and Tate 2012) and status as CEO appointees (Coles, Daniel, and Naveen 2014) affect board monitoring.

³ For example, Exelon Corporation established a \$50 million foundation in the fourth quarter of 2007, right before the financial market crisis. Exelon's chairman and CEO, John Rowe, served as the chairman and director of the foundation from its inception to 2012 (the end of our sample period). Consistent with our view, Exelon routinely donated to multiple charities affiliated with each of its independent directors (and, in several cases, their spouses). For example, in 2008 alone, seven charities affiliated with the chairwoman of the compensation committee (Rosemarie Greco) received donations from Exelon, as did five charities affiliated with its lead independent director, who was also a member of the compensation committee (Walter D'Alessio). "All payments were immaterial under the relevant independence criteria," stated Exelon in its 2008 proxy filing. Exelon Foundation funded some of the contributions to independent director-affiliated charities. A subsidiary of Exelon, PECO, funded other affiliated donations.

⁴ According to NYSE rules, only corporate donations that exceed the greater of \$1 million or 2% of the consolidated gross revenue of a charity in which an independent director serves as an executive officer in any of the past three years are required to be disclosed in the typical data sources for corporate governance research (i.e., on the company's website, in its annual proxy statement, or in its 10-K filing). All corporate donations through their

largely overlooked in corporate governance research until very recently. In this study, we test whether CEOs gain leverage over independent directors by channeling corporate donations to directors' affiliated charities. We examine the determinants of affiliated donations and their effect on board monitoring. We find that firms with weaker board oversight and shareholder monitoring are more likely to make affiliated donations, and such donations are larger. Moreover, long-tenured directors and directors affiliated with more charities are more likely to receive corporate donations. Interestingly, affiliated donations typically begin after an independent director's appointment to the board that connects the firm and the charity, and they end after his or her exogenous departure that severs the connection. This indicates that affiliated donations target independent directors rather than particular charitable causes.

To examine the effects of affiliated donations, we focus primarily on CEO compensation practices; we also study CEO replacement decisions and firm performance. Greater CEO compensation and retention of a poorly performing CEO are often signs of governance failures (Weisbach 1988; Parrino 1997; Core et al. 1999; Faleye, Hoitash, and Hoitash 2011). We find that firms making affiliated donations pay their CEOs 9.4% more on average than firms not making affiliated donations, after controlling for common determinants of CEO compensation. To identify the channel by which affiliated donations affect CEO compensation, we contrast donations made to charities affiliated with independent directors serving on the compensation committee with donations made to charities involving other independent directors. We show that the positive association between affiliated donations and the level of CEO compensation is significant only when affiliated donations involve compensation committee members. On average, at firms that donate to charities affiliated with the compensation committee chair (a large fraction of the compensation committee), CEO compensation is 15.6% (13.7%) higher than it is at firms that do not make affiliated donations.

Endogeneity is an important concern in studies on corporate governance. One potential problem is that the positive association between affiliated donations and CEO compensation may be driven by omitted variables beyond the firm, CEO, and governance characteristics included in our analyses. We conduct a battery of tests to address such concerns: examining changes in CEO compensation around the initiation and termination of affiliated donations; including firm fixed effects, firm-by-CEO fixed effects, and director fixed effects in the analysis; and using the propensity score matching (PSM) approach. We further control for alternative channels such as unaffiliated donations, CEO-affiliated donations, inside director-affiliated donations, CEO-director ties at affiliated charities, and free cash flow (Jensen 1986).

private foundations, regardless of the amount, are included in Form 990-PF, filed annually with the IRS. This is the original source of our data.

The positive correlation between affiliated donations and CEO compensation continues to hold in all the tests.

Because it is impossible to exhaust all omitted variables that may drive the positive correlation between affiliated donations and CEO compensation, we construct an instrumental variable based on an exogenous shock to the demand for affiliated donations. More specifically, we identify forced turnovers of college basketball and football head coaches. We focus on these events because over 40% of affiliated donations go to educational institutions and forced turnovers of college head coaches are very common—and often costly. Firing a coach for poor performance and revamping the team can cost millions of dollars (for an example of forced coach turnover, see Appendix A). Head coach terminations tend to happen after a prolonged period of poor performance, when advertisement revenue and alumni donations are below expectations. Under pressure to fundraise, independent directors who are affiliated with the universities greatly appreciate CEOs' understanding and corporate contributions at these critical moments, and this appreciation may weaken their monitoring incentive. Importantly, college coach replacement is unlikely to be related to CEO compensation. It affects CEO compensation only via a firm's decision to make affiliated donations in response to the increased demand for funding. Our instrumental variable regression shows that CEO compensation increases with affiliated donations predicted by forced turnovers of college coaches. This result directly supports our interpretation that affiliated donations affect CEO compensation.

In addition to their effect on CEO compensation, we examine how affiliated donations affect CEO replacement decisions, another important monitoring task performed by boards (Weisbach 1988; Parrino 1997; Kaplan and Minton 2012; Jenter and Kanaan 2015; Gao, Harford, and Li 2017). Poor firm performance is often used as a proxy for low CEO ability. Retaining a weak (i.e., low-ability) CEO can be costly to shareholders. We show that affiliated donations attenuate the link between forced CEO turnover and firm performance: a CEO is unlikely to be replaced for poor performance if the firm donates to charities affiliated with a large fraction of the board (i.e., greater than the sample median) or if the firm makes large affiliated donations. Lastly, if affiliated donations weaken board monitoring effectiveness, we expect firm performance and shareholder value to be negatively affected. We test this prediction using various measures of stock and accounting performance and show that this is indeed the case.

This study uncovers an important channel through which independent directors' monitoring incentives can be compromised (Core et al. 1999; Faleye et al. 2011). Affiliated donations affect board monitoring after we account for CEO–director social connections (Hwang and Kim 2009; Fracassi and Tate 2012) and directors' status as CEO appointees (Coles et al. 2014). In contrast to static social connections and CEO–appointee status, a material relationship between independent directors and the CEO via affiliated donations evolves over time. This relationship is similar to that created by interlocking

directorships (Hallock 1997; Fich and White 2003), and it is built at the expense of taxpayers and shareholders. Our findings suggest that regulators should mandate disclosure of affiliated donations in SEC filings (e.g., in firms' proxy statements) that are easily accessible to investors. Such disclosure would help inform shareholders of independent directors' potential conflicts of interest, leading to a more accurate definition of director independence.

In a related study, Yermack (1996) documents large charitable stock gifts made by chairmen and CEOs to their family foundations just before sharp declines in stock prices. Yermack attributes such behavior to the CEOs' fraudulent backdating of stock gifts to increase personal income tax benefits. A separate study by Masulis and Reza (2015) focuses on the effect of CEO charity preferences on corporate giving and concludes that corporate giving reflects an agency problem. Masulis and Reza (2015) provide auxiliary evidence suggesting that CEO compensation is higher at firms whose major charitable causes overlap with those of their independent directors (based on broadly defined charity types, such as education and health centers). Such overlaps are crude proxies for potential connections between the CEO and independent directors through charitable donations. In contrast, we identify corporate donations actually made to charities affiliated with independent directors by matching the recipients of a company's charitable contributions with the names of director-affiliated charities. Such matches enable us to show that corporate donations tend to start after an independent director's appointment to the board that establishes a firm–charity connection and end following his or her exogenous departure that severs the connection. We also examine how director traits affect affiliated donations, investigate how affiliated donations affect CEO compensation through compensation committee members, and identify a causal effect of affiliated donations on the effectiveness of board monitoring.

Firms could camouflage affiliated donations as corporate social responsibility (CSR) investments, thereby underestimating director compensation costs and hiding the material relationship between affiliated independent directors and the CEO. This possibility ties our paper to the growing literature examining whether corporate spending on CSR improves firm value (Edmans 2011; Flammer 2015) or satisfies CEOs' personal preferences to the detriment of shareholders (Friedman 1970; Cheng, Hong, and Shue 2016).⁵ However, the CSR literature does not examine how CEOs benefit personally from building relationships

⁵ Empirical evidence in the CSR literature is mixed. Edmans (2011) shows that corporate goodness improves employee morale and efficiency, and Flammer (2015) shows that the adoption of CSR-related shareholder proposals that narrowly passed the majority vote leads to positive announcement returns and superior accounting performance via increases in labor productivity and sales growth. In contrast, Cheng et al. (2016) state that corporate goodness reflects an agency problem because it is negatively associated with governance strength and after-tax insider ownership, and they show that the passage of shareholder governance proposals leads to slower growth in corporate goodness. Friedman (1970) writes, "There is one and only one social responsibility of business—to use its resources and engage in activities designed to increase its profits so long as it stays within the rules of the game."

with independent directors by channeling corporate funds to director-affiliated charities.⁶

The remainder of the paper is organized as follows. Section 1 describes our conceptual framework and develops empirical hypotheses. Section 2 describes our data. Section 3 examines the economic determinants of affiliated donations. Section 4 uncovers the link between affiliated donations and CEO compensation. Section 5 addresses endogeneity concerns and rules out alternative hypotheses. Section 6 further examines how affiliated donations are related to firm performance and the sensitivity of forced CEO turnovers to performance. Section 7 concludes.

1. Conceptual Framework and Hypothesis Development

Director independence is an essential element of corporate governance, which refers to the ways in which capital suppliers seek to ensure that their capital is wisely managed (Hermalin and Weisbach 2017). Corporate governance contains a complex set of interrelated aspects: (1) takeovers and other market forces (Giroud and Mueller 2010), (2) ownership structure (ownership concentration, director ownership, and CEO ownership) and shareholder rights (engagement vs. voting with your feet), and (3) the board of directors. The board of directors is tasked with performing the critical functions of monitoring and advising management. Monitoring mitigates the conflicts of interest between managers and shareholders, such as the free cash flow problem. Its effectiveness depends on independent directors' incentive and ability. In this paper, we examine whether CEOs gain leverage over conventionally independent directors by channeling corporate donations to directors' affiliated charities, ultimately reducing their monitoring incentive.

One of the major responsibilities of nonprofit directors, officers, and trustees is fundraising.⁷ Corporate donations help directors fulfill fundraising obligations at their affiliated charities. This, in turn, enables the director to achieve personal philanthropic ambitions, whether they involve pursuing a personal passion or achieving social prominence. We can thus view affiliated donations as hidden payments to independent directors, coming on top of the highly visible direct compensation received for their board services.

⁶ Our paper is also related to the literature examining the relationship between political connections and firm value (Fisman 2001; Faccio 2006; Goldman, Rocholl, and So 2009; Cooper, Gulen, and Ovtchinnikov 2010; Aggarwal, Meschke, and Wang 2012; Coates 2012; Schoenherr 2019). Bertrand et al. (2018) examine corporate giving with a political agenda. They find that corporate giving to charities located in congressional districts increases when a representative obtains seats on committees of policy relevance to the firm, and that corporate giving to such charities declines when the representative leaves office. They conclude that corporate giving is a form of tax-exempt influence-seeking that remains mostly undetected due to lack of formal disclosure. However, the research that links political connections to firm value does not examine how such connections affect the bargaining power between CEOs and independent directors.

⁷ See, for example, the website of the Georgia Center for Nonprofits: <http://www.gcn.org/articles/what-role-do-boards-and-individual-board-members-have-in-nonprofit-fundraising#sthash.SQtSPOaU.dpuf>.

Because CEOs often direct corporate charitable giving, independent directors whose charities receive corporate donations have weakened bargaining power over CEOs (Hermalin and Weisbach 1998). This changes a director's status from independent to dependent, posing a conflict of interest in representing shareholders.

In establishing our empirical hypotheses, we focus on the effect of affiliated donations on CEO compensation, an important task for boards of directors. While executive compensation should be determined by economic factors—such as managerial ability, job responsibility, and business risks—such practices can be hampered by agency problems. Weak corporate governance structures can lead boards to award CEOs greater excess pay (CEO compensation in excess of the level predicted by economic determinants) to the detriment of shareholders (Jensen 1993; Core et al. 1999; Bertrand and Mullainathan 2001; Bertrand et al. 2018; Faulkender and Yang 2010; Faleye et al. 2011). We examine whether firms making affiliated donations award more generous CEO pay.

Boards sometimes have to replace poorly performing CEOs (Weisbach 1988; Parrino 1997; Kaplan and Minton 2012; Jenter and Kanaan 2015; Gao, Harford, and Li 2017). If an independent director's affiliated charities receive corporate donations, the director may be less willing to support a needed CEO transition.⁸ To retain a CEO with a poor track record, a sufficient number of independent directors must be in agreement. We posit that the sensitivity of forced CEO turnover to firm performance is lower if a firm makes intensive affiliated donations (i.e., donating to charities affiliated with a large fraction of independent directors or donating large amounts). Moreover, if affiliated donations weaken a board's monitoring effectiveness, firm performance and shareholder value should suffer. We test this corollary using four stock and accounting performance measures.

The observed link between affiliated donations and CEO compensation could be driven by other factors. First, it is possible that any corporate donations, affiliated or not, reflect agency problems (Masulis and Reza 2015). We contrast the effect of affiliated donations on CEO compensation with that of corporate donations to charities not affiliated with independent directors (*unaffiliated donations*). Second, it is possible that affiliated donations coincide with corporate donations to charities also affiliated with the CEO or other inside directors (e.g., independent directors and inside directors who serve on the board of the same charity). Thus, we test the effect of affiliated donations on CEO compensation, controlling for CEO-affiliated donations (Masulis and Reza 2015) and inside director-affiliated donations, respectively. Third, affiliated donations may simply reflect social ties between independent directors

⁸ Beneish, Marshall, and Yang (2017) show that independent directors who benefit by actively selling stock contemporaneously with the CEO during a misreporting period are more likely to be aligned with the CEO and less likely to fire him or her after the discovery of the fraud.

and the CEO. To address this concern, we further control for whether the CEO is connected to independent directors via the recipient charity. Fourth, affiliated donations may reflect other corporate governance issues, such as the free cash flow problem. In additional tests, we control for various measures of free cash flow when examining the link between affiliated donations and CEO compensation.

Different corporate governance mechanisms often interact with one another. The CEO's ability to gain leverage over independent directors can be mitigated by other governance forces, such as strong shareholder rights. Acknowledging this possibility, we control for other corporate governance mechanisms in establishing a causal effect of affiliated donations on board monitoring effectiveness.

2. Data Description and Univariate Analysis

Our sample consists of the S&P 500 index companies as of the end of 2012. We extract director data from the BoardEx database, which includes board committee assignments and a director's affiliated tax-exempt organizations (*charities*). We obtain CEO compensation information from ExecuComp, firm financial and market data from Compustat and CRSP, and governance variables from Riskmetrics and Thomson Reuters.

We extract data on corporate charitable donations from *Foundation Directory Online* (FDO) at <https://fconline.foundationcenter.org/>. FDO data start in 2003 and are compiled by Foundation Center from IRS Forms 990-PF (Return of Private Foundations) and 990 (Return of Organization Exempt from Income Tax), grant maker Web sites, annual reports, printed application guidelines, the philanthropic press, and various other sources. Generally, all grants of more than \$10,000 are included for all foundations with a total giving amount of at least \$5 million (roughly, the top 1,000 donors) in a given year.⁹ The database also contains charitable donations via corporate giving programs, typically for natural disasters. We match the directors' affiliated charities listed in BoardEx with charities that received corporate donations using charity names.¹⁰ Starting from S&P 500 firms, we lose ten firms with missing BoardEx data, twenty-two foreign firms, four firms with missing stock market variables, and twenty

⁹ In addition, an assortment of grants of less than \$10,000 is included for these foundations, and an assortment of grants of all sizes is included for foundations with a total giving amount of less than \$5 million. Some of the data are generated in response to custom requests by database users for information on donations to certain types of charities. The database keeps corporate donations of at least \$1,000, which helps us filter out corporate matching programs for employee donations. Our results are robust whether we include donations below \$10,000 or not (untabulated).

¹⁰ We use the Python function *ratio* to match names. The initial sample comprises perfect matches returned by running *ratio*. We supplement the data by manually checking all potential matches with a matching score greater than 0.85.

firms with missing governance variables. Our final sample contains 3,746 firm-years from 444 unique S&P 500 firms that are nonforeign and have all of the information required for our analyses from 2003 to 2012.

Table 1 includes the summary statistics of the donation data. Panel A compares affiliated donations with all donations, including both affiliated and unaffiliated donations. It shows that 1,398 out of 3,746 firm-year observations (37.3%) in our sample make charitable contributions (of at least \$5 million in total giving), and the average amount is \$11.1 million provided that a donation is made. Approximately two-thirds of the donating firm-years (917 firm-year observations, 24.5% of the sample) contribute to charities affiliated with their independent directors.¹¹ The average affiliated donation amount for each firm-year is \$1.5 million, and these donations are affiliated with three independent directors on average. Panel B shows the occurrence and amount of affiliated donations by year. The highest level occurs in 2003. Telephone and television transmission, finance, and manufacturing are the three industries in which firms are most likely to make affiliated donations based on the number of firm-year observations. Manufacturing, consumer durables, and finance are the top three industries based on the number of firms making affiliated donations (panel C).

We report summary statistics on CEO compensation, corporate governance, firm financials, and director characteristics in Table 2. Panel A compares firms that make affiliated donations with firms that do not. Panel B compares directors whose affiliated charities have received one or more donations from the firms on whose boards they serve with those whose charities have not received any such donations. Firms making affiliated donations differ from other firms along various dimensions of corporate governance. Specifically, CEOs of donating firms receive greater total annual pay. They are more likely to serve as the chairman of the board, have a shorter tenure as CEO, and own a smaller fraction of the firm's equity. Not surprisingly, when non-co-opted independent directors (independent directors who join the board before the CEO takes the corner office) are affiliated with more charities, firms are more likely to make affiliated donations. A CEO whose appointment was approved by these longer-tenured independent directors has a stronger incentive to gain leverage over the directors via charitable donations to their affiliated charities.

Firms making affiliated donations tend to have larger boards, more social connections between independent directors and the CEO, and lower ownership by the top five institutional investors. Overall, firms with weak corporate governance are more likely to make affiliated donations. Moreover, we show that firms making affiliated donations tend to be larger (by book value of assets), have inferior stock performance (lower stock returns and market-to-book ratio),

¹¹ In our sample, thirty-four firms make affiliated donations throughout their years in the sample; sixty-three firms make affiliated donations in more than half of but not all their years in the sample; eighty-six firms make affiliated donations in less than half of their years in the sample. The remaining 261 firms never make affiliated donations during their years in the sample.

Table 1
Summary statistics of affiliated donation variables

Panel A: Affiliated donations vs. general donations

	Full sample			
	No. of firm-years	Mean	Median	Std. dev.
Decision to make donation:				
All donations	3,746	0.373	0	0.484
Affiliated donations	3,746	0.245	0	0.430
Amount of donation, provided that a donation is made (in millions of dollars):				
All donations	1,398	11.131	4.262	21.878
Affiliated donations	917	1.498	0.429	3.010

Panel B: Distribution by year and summary statistics

Year	No. of firms	No. of firms with affiliated donations	Amount of affiliated donation provided affiliated donation is made (\$) (in dollars)			
			Mean	p25	Median	p75
2003	311	74	1,962,228	150,500	731,554	1,528,240
2004	316	79	1,274,669	107,000	512,447	1,775,000
2005	327	93	1,381,310	54,000	440,000	1,545,000
2006	356	99	1,226,279	51,200	459,000	1,296,400
2007	374	106	1,120,274	57,583	285,095	1,250,000
2008	382	95	1,664,540	100,000	330,000	1,633,250
2009	406	98	1,352,304	50,000	385,000	1,216,213
2010	415	103	1,689,392	80,000	343,000	1,440,600
2011	427	95	1,689,790	69,501	454,048	1,599,520
2012	432	75	1,787,140	99,140	451,030	1,188,000
Total	3,746	917	1,498,088	75,000	429,000	1,464,706

Panel C: Distribution by industry

Industry	No. of firm-years	No. of firm-years with affiliated donations	%	No. of firms	No. of firms with affiliated donations	%
Telephone and television transmission	78	33	42	15	5	33
Finance	595	221	37	76	41	54
Manufacturing	392	140	36	43	25	58
Consumer nondurables	259	78	30	34	14	41
Consumer durables	70	20	29	7	4	57
Health care, medical equipment, and drugs	276	75	27	32	14	44
Chemicals and allied products	149	39	26	17	8	47
Utilities	280	72	26	32	17	53
Other	384	79	21	43	15	35
Wholesale, retail, and some services	418	84	20	48	17	35
Oil, gas, and coal extraction and products	235	32	14	27	6	22
Business equipment	610	44	7	70	17	24
Total	3,746	917	24	444	183	41

The sample consists of S&P 500 firms from 2003 to 2012 that have the information required for our analyses. Panel A provides summary statistics on affiliated donations and on all donations (affiliated and unaffiliated). Panel B tabulates the number of firms in the sample, the number of firms making affiliated donations, and the dollar amount of affiliated donations by year. Panel C tabulates the distribution by industry.

appear less risky (lower stock return volatility and research and development [R&D] investments), and are located in states with higher corporate tax rates than firms not making such donations.

Table 2
CEO compensation and firm characteristics by whether affiliated donation is made

Panel A: Firm-level variables

Affiliated donation is made:	Yes			No			Yes minus No	
	N	Mean	Std. dev.	N	Mean	Std. dev.	Diff	t-stat
<i>CEO compensation</i>								
Ln(Total CEO pay in thousands of dollars)	917	9.130	0.701	2,829	8.792	0.778	0.338	12.36
<i>Corporate governance</i>								
D(CEO serving as chairman)	917	0.470	0.499	2,829	0.345	0.475	0.125	6.68
CEO tenure	917	5.194	4.438	2,829	6.434	5.896	-1.240	-6.75
CEO ownership (%)	917	0.247	0.812	2,829	0.913	2.741	-0.667	-11.48
Ln(No. of charities affiliated with non-co-opted independent directors)	917	3.433	1.625	2,829	2.581	1.735	0.852	13.57
Board size	917	11.944	2.266	2,829	10.198	2.161	1.746	20.51
D(Independent board)	917	0.924	0.266	2,829	0.915	0.279	0.009	0.87
D(Social connection(s) between independent directors and CEO)	917	0.457	0.498	2,829	0.345	0.475	0.112	5.98
Top five institutions' ownership	917	0.231	0.072	2,829	0.263	0.085	-0.032	-11.08
<i>Firm financials</i>								
Ln(Assets)	917	9.932	0.926	2,829	9.127	1.089	0.805	21.88
Stock return	917	0.116	0.319	2,829	0.157	0.344	-0.041	-3.35
Profit margin	917	0.232	0.129	2,829	0.229	0.143	0.003	0.66
Market-to-book ratio	917	1.775	0.880	2,829	2.110	1.316	-0.335	-8.79
Stock return volatility	917	0.315	0.192	2,829	0.340	0.176	-0.025	-3.48
Debt/assets	917	0.237	0.142	2,829	0.227	0.168	0.010	1.74
Advertisement/sales	917	0.012	0.024	2,829	0.010	0.025	0.001	1.33
R&D/assets	917	0.019	0.034	2,829	0.025	0.053	-0.007	-4.47
Industry growth	917	0.082	0.089	2,829	0.079	0.088	0.003	0.83
State corporate tax rate	917	0.069	0.028	2,829	0.065	0.030	0.004	3.38

(Continued)

Directors whose affiliated charities receive corporate donations tend to have more charity affiliations, hold more board seats at other S&P 500 firms, are more likely to be a named executive officer (NEO) at another S&P 500 firm, and are more likely to have social connections to the CEO. These affiliated directors also tend to be older, have longer tenure relative to the CEO, are less likely to be the chair of a board committee, and are more likely to be female. Additionally, affiliated directors receive higher compensation on average than other directors but are less likely to be in the top decile of director compensation. Finally, they are less likely to have prior industry experience.

Corporations usually maintain long-lasting relationships with affiliated charities and broadly donate to similar types of charities over time.¹² The nature of donating firms' business tends to match the mission of the recipient

¹² Our sample includes 894 unique charitable organizations that receive affiliated donations, corresponding to 4,774 firm-year-charity observations. We find that, on average, a charity receives affiliated donations from a particular corporate giver for three consecutive years. Conditional on the firm making affiliated donations to a particular charity, 59.5%, 41.9%, and 32.3% continue to make donations to the charity in the following one year, three years, and five years, respectively. We categorize the 4,774 charities into five groups: 41.7% are educational institutions, 5.4% are health-related organizations, 12.9% are arts and cultural organizations, 7.3% are disaster relief organizations, and 32.7% are miscellaneous philanthropic organizations. We find that among firms that make affiliated donations in our sample, 80.9%, 70%, and 66% keep the same donation pattern for four or five types of charities in the following one year, three years, and five years, respectively.

Table 2
(Continued)

Panel B: Director characteristics

Variable	Yes			No			Yes minus No	
	<i>N</i>	Mean	Std. dev.	<i>N</i>	Mean	Std. dev.	Diff	<i>t</i> -stat
Affiliated donation is made:								
Ln(No. of charities director is affiliated with)	2,391	2.645	0.602	23,150	1.915	0.916	0.729	53.23
Ln(No. of S&P 500 boards director sits on)	2,391	0.392	0.457	23,150	0.263	0.404	0.129	13.29
D(Director is an NEO at another S&P 500 firm)	2,391	0.058	0.233	23,150	0.049	0.215	0.009	1.82
D(Social connection(s) between independent directors and CEO)	2,391	0.127	0.333	23,150	0.088	0.283	0.039	5.57
D(Director is independent chairman)	2,391	0.023	0.150	23,150	0.027	0.162	-0.004	-1.23
D(Director is lead director)	2,391	0.088	0.284	23,150	0.089	0.285	-0.001	-0.14
D(Director is chair of a board committee)	2,391	0.299	0.458	23,150	0.317	0.465	-0.017	-1.77
Ln(1+director pay in \$1,000)	2,391	5.350	0.493	23,150	5.309	0.579	0.041	3.84
D(Director pay is in the top decile)	2,391	0.079	0.271	23,150	0.103	0.304	-0.023	-3.96
D(Director has prior industry experience)	2,391	0.131	0.338	23,150	0.182	0.386	-0.051	-6.90
Director age	2,391	57.065	4.919	23,150	56.369	5.965	0.696	6.45
Director tenure in excess of CEO tenure	2,391	9.264	6.896	23,150	8.146	7.011	1.118	7.54
D(Director is female)	2,391	0.206	0.404	23,150	0.176	0.381	0.029	3.39

The sample consists of S&P 500 firms from 2003 to 2012 that had available the information required for our regression analyses. Panel A compares CEO compensation, corporate governance, and firm financials between firm-years with affiliated donations and those without such donations. Panel B compares director characteristics between director-years with affiliated donations and those without such donations. All variables are described in Appendix B. All continuous variables are winsorized at the first and ninety-ninth percentiles.

charities. For example, high-tech firms have more independent directors affiliated with universities, and pharmaceutical firms have more independent directors affiliated with health and medical organizations. A significant portion of affiliated donations goes to local charities: among our sample of charities that receive affiliated donations, 26.6% are located in the same city as the donating firm’s headquarters.¹³

3. Determinants of Affiliated Donations

3.1 Determinants of affiliated donations in a level specification

We next examine the economic determinants of corporate donations to charities affiliated with independent directors. We focus on the relationship between affiliated donations and corporate governance, and we use a regression specification similar to that used in Masulis and Reza (2015). If affiliated

¹³ To check whether the matching between donating firms and recipient charities confounds our results, we include in our (firm-year-charity level) regression of CEO total pay on affiliated donations the matching types of donations (e.g., the business equipment industry matched with educational organizations, and the health care industry matched with health and medical organizations) and local donations. We report in the Online Appendix that neither of these indicators is significantly related to CEO pay, and affiliated donations remain an economically and statistically significant determinant of CEO pay.

donations serve mainly as side payments to independent directors, they should be more likely to occur and larger when corporate governance is weaker. For comparison, we also analyze the economic determinants of unaffiliated donations.¹⁴ We include firm financial characteristics in our analyses as control variables. We further include industry fixed effects to parse out differences in donations across industries and year fixed effects to control for general time trends in corporate donations. We cluster standard errors at the firm level to account for serial correlations in the data.

We show that firms with lower CEO ownership and a larger board are more likely to make affiliated donations and to donate larger amounts (panel A of Table 3). Consistent with our univariate results, if long-tenured independent directors have more charity affiliations, firms are more likely to donate to these charities and make larger donations. In addition, firms are more likely to make affiliated donations and donate more when independent directors have social connections to the CEO and when the top five institutional investors own a smaller percentage of the company. These results suggest that large affiliated donations are more likely to occur at firms with weak monitoring by directors and large shareholders and at firms where the interests of the CEO and the shareholders are less closely aligned. In contrast, governance measures do not affect the size of unaffiliated donations. The likelihood of unaffiliated donations correlates only with CEO ownership, and this negative correlation is marginal. The comparison of donation determinants reinforces our view that the problematic relationship between charitable donations and corporate governance pertains mainly to affiliated donations.

Regarding firm characteristics, we find that firm size, market-to-book ratio, R&D investment, and industry growth are positively correlated with the likelihood and amount of affiliated donations. In addition, the amount of affiliated donations increases with financial leverage. In contrast, only firm size is positively correlated with the likelihood and amount of unaffiliated donations, perhaps because large firms have more resources for charitable causes.

We next investigate which director characteristics affect the likelihood of receiving affiliated donations. We include thirteen director characteristics in our regression specifications and conduct the analysis at the independent director-year level. A well-connected director is presumably more valuable to the firm and should be paid more, directly or indirectly. We use four director connection measures: the number of the director's affiliated charities, the number of S&P 500 boards on which the director serves, a dummy variable indicating whether

¹⁴ Throughout the paper, the term "unaffiliated donations" is defined as corporate donations made to charities not affiliated with independent directors. Unaffiliated donations include contributions to charities affiliated with the CEO or other top executives but not affiliated with independent directors of the firm. In unreported analyses, we use a multinomial regression model with three categories: (1) firms that make no donations, (2) firms that make unaffiliated donations only, and (3) firms that make affiliated donations (in all but two cases, unaffiliated donations are also made in years when the same firm made affiliated donations). Our results on how corporate governance and firm characteristics affect corporate donations are very similar.

Table 3
Determinants of affiliated and unaffiliated donations
Panel A: Firm-level regressions of affiliated and unaffiliated donations

Dependent variable	Affiliated donation		Unaffiliated donation	
	Propensity (1)	Amount (2)	Propensity (3)	Amount (4)
<i>Corporate governance</i>				
D(CEO serving as chairman)	0.071** (2.24)	0.505** (1.96)	0.017 (0.83)	0.877 (1.17)
CEO tenure	0.001 (0.31)	0.011 (0.55)	-0.002 (-1.44)	-0.060 (-1.06)
CEO ownership	-0.024*** (-2.90)	-0.203*** (-2.84)	-0.007* (-1.86)	-0.181 (-1.46)
Ln(No. of charities affiliated with non-co-opted independent directors)	0.030*** (3.42)	0.295*** (3.90)	-0.007 (-1.45)	-0.244 (-1.39)
Board size	0.028*** (4.57)	0.201*** (3.65)	-0.000 (-0.10)	-0.011 (-0.07)
D(Independent board)	0.015 (0.34)	0.163 (0.47)	0.030 (1.20)	1.450 (1.30)
D(Social connection(s) between independent directors and CEO)	0.047* (1.86)	0.517** (2.40)	0.009 (0.60)	0.098 (0.18)
Top five institutions' ownership	-0.303** (-2.01)	-4.379*** (-3.28)	-0.133 (-1.38)	-4.946 (-1.26)
<i>Firm financials</i>				
Ln(Assets)	0.093*** (5.07)	0.920*** (5.61)	0.025** (2.38)	1.305** (2.29)
Stock return	-0.012 (-0.49)	-0.070 (-0.38)	-0.008 (-0.36)	-0.857 (-1.05)
Profit margin	-0.047 (-0.41)	-1.431 (-1.30)	-0.019 (-0.29)	0.897 (0.35)
Market-to-book ratio	0.023* (1.68)	0.269** (2.40)	0.003 (0.25)	0.248 (0.63)
Stock return volatility	-0.006 (-0.07)	-0.296 (-0.44)	-0.089 (-1.20)	-2.854 (-0.93)
Debt/assets	0.017 (0.20)	1.477* (1.90)	0.015 (0.28)	0.157 (0.08)
Advertisement/sales	0.010 (0.02)	1.397 (0.29)	0.228 (0.66)	14.814 (0.98)
R&D/assets	0.701** (2.12)	6.307** (2.24)	-0.057 (-0.26)	2.365 (0.30)
Industry growth	0.202*** (2.97)	1.231*** (2.68)	-0.039 (-0.60)	-0.757 (-0.30)
State corporate tax rate	0.354 (0.69)	-0.457 (-0.11)	0.521 (1.60)	7.454 (0.61)
Industry FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
No. of observations	3,746	3,746	3,746	3,746
Pseudo R^2	.209	.140	.069	.037

(Continued)

the director is a top executive of another S&P 500 firm, and a dummy variable indicating whether the director and CEO have social connections. We also use three dummy variables that measure the importance of a director's role on the board, noting whether a director is the chairman of the board, the lead independent director, or the chair of a board committee. We further include the total annual compensation that an independent director receives from the firm and a dummy variable indicating whether this compensation is in the top decile

Table 3
(Continued)

Panel B: Director-level regressions of affiliated donations

Dependent variable	Propensity to donate (1)	Donation amount (2)
<i>Director characteristics</i>		
Ln(No. of charities director is affiliated with)	0.045*** (9.39)	0.660*** (6.43)
Ln(No. of S&P 500 boards director sits on)	0.006 (1.18)	0.041 (0.51)
D(Director is an NEO at another S&P 500 firm)	-0.007 (-1.06)	-0.108 (-0.92)
D(Social connection(s) between independent director and CEO)	0.003 (0.40)	0.070 (0.60)
D(Director is independent chairman)	0.018 (0.95)	0.178 (0.90)
D(Director is lead director)	0.001 (0.11)	-0.006 (-0.05)
D(Director is chair of a board committee)	-0.002 (-0.70)	-0.007 (-0.14)
Ln(1+director pay in \$1,000)	0.001 (0.20)	0.003 (0.03)
D(Director pay is in the top decile)	-0.017** (-2.27)	-0.211 (-1.60)
D(Director has prior industry experience)	-0.010 (-1.36)	-0.082 (-0.65)
Director age	0.000 (0.12)	0.004 (0.32)
Director tenure in excess of CEO Tenure	0.001*** (4.51)	0.021*** (3.81)
D(Director is female)	-0.003 (-0.65)	-0.100 (-1.55)
<i>Corporate governance</i>		
D(CEO serving as chairman)	0.037*** (3.10)	0.374*** (2.74)
CEO tenure	-0.001 (-0.73)	-0.008 (-0.61)
CEO ownership	-0.010*** (-2.76)	-0.149** (-2.51)
Board size	0.005*** (2.62)	0.056** (2.00)
D(Independent board)	-0.020 (-0.82)	-0.205 (-0.86)
Top five institutions' ownership	-0.165*** (-3.07)	-2.686*** (-3.16)
<i>Firm financials</i>		
Ln(Assets)	0.025*** (4.35)	0.376*** (4.37)
Stock return	0.002 (0.27)	-0.022 (-0.27)
Profit margin	0.004 (0.13)	-0.171 (-0.39)
Market-to-book ratio	0.003 (0.59)	0.052 (0.66)
Stock return volatility	0.013 (0.62)	0.105 (0.39)
Debt/assets	0.043* (1.74)	0.805** (2.23)
Advertisement/sales	-0.044 (-0.25)	0.066 (0.03)

(Continued)

Table 3
(Continued)

Panel B: Director-level regressions of affiliated donations

Dependent variable	Propensity to donate (1)	Donation amount (2)
R&D/assets	0.223** (2.28)	2.932** (2.12)
Industry growth	0.039*** (2.55)	0.357* (1.85)
State corporate tax rate	0.086 (0.54)	0.754 (0.35)
Industry FEs	Yes	Yes
Year FEs	Yes	Yes
No. of observations	25,541	25,541
Pseudo R ²	.212	.165

This table examines the determinants of affiliated and unaffiliated donations. In panel A, the propensity to make affiliated donations equals one if a firm makes affiliated donations in a given year and zero otherwise. The propensity to make unaffiliated donations equals one if a firm makes unaffiliated but not affiliated donations in a given year and zero otherwise. The amount of affiliated donations is the total size of affiliated donations made over the course of the year. The amount of unaffiliated donations is the total size of donations made over the course of the year excluding affiliated donations. The data used in panel B are at the director-year level. The propensity to donate equals one if the company donates to an independent director's affiliated charities and zero otherwise. The donation amount is the total amount of all donations made to charities affiliated with the director. Regressions of the donation propensity use the probit model, and regressions of the donation amount use the Tobit model. Marginal effects of the coefficients are reported. All regressions include industry fixed effects and year fixed effects. Industries are Fama-French 12 industries. All variables are described in Appendix B. We report *t*-statistics based on heteroscedasticity-robust standard errors adjusted for firm clusters in parentheses below the corresponding regression coefficients. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Abbreviation: FE, fixed effects.

of our sample.¹⁵ Because greater compensation may indicate higher aptitude, we use director pay measures to test the possibility that firms compensate the most talented directors by donating to their charities in addition to awarding them generous pay. Lastly, we consider a director's industry experience, tenure (as compared with CEO tenure), age, and gender.

In panel B of Table 3, we show that the likelihood and size of affiliated donations are greater for independent directors who are affiliated with more charities and for those who have been serving on the board longer than the CEO has been in the corner office. Both findings suggest that CEOs tend to use charitable giving to gain bargaining power over influential independent directors. We find little evidence that affiliated donations complement annual director compensation—director pay is not correlated with affiliated donations. The likelihood of receiving affiliated donations is lower if the independent director's pay is in the top decile.

The effects of corporate governance and firm financials on the likelihood and size of affiliated donations are similar to the results presented in panel A. For example, we find that firms with weaker corporate governance (CEO serves as chairman, lower CEO ownership, lower ownership by the top five

¹⁵ Compensation data for individual directors became available in 2006, so our sample period for panel B of Table 3 is from 2006 to 2012. Our results remain qualitatively the same if we complete the sample by using the average director pay across each firm as a proxy for individual director pay for 2003 through 2005.

institutional investors, and a larger board) are more likely to donate and make larger donations to charities affiliated with independent directors.

3.2 Changes in affiliated donations around directors' appointments and exogenous departures

If affiliated donations are intended to help the CEO gain leverage over the board, we expect the initiation (termination) of affiliated donations to follow independent directors' appointments (departures). We predict that when an independent director joins the board and establishes the first connection between a firm and a charity (i.e., no incumbent directors are connected to the charity at the time), the firm is more likely to initiate donations to his or her affiliated charity than to initiate donations to charities affiliated with an incumbent.¹⁶ Conversely, a firm is more likely to terminate donations to charities affiliated with an independent director when the director leaves the firm for exogenous reasons and the departure eliminates the firm's connection to the charity (i.e., no directors are connected to the charity after the event) than it is to terminate donations to charities affiliated with the remaining directors. We test these two predictions at the firm-director-year-charity level.

Exogenous director departure is measured in three different ways. The first measure is director retirement, defined as a departure at the age of seventy or older (Stern 2017). Results are robust to using seventy-two or sixty-five as the retirement age. The second measure is sudden departure, defined as a departure followed by departures from all board positions within two years.¹⁷ The third measure of exogenous director departure is the union of the first two measures. By focusing on exogenous departures, we minimize the possibility that the termination of affiliated donations is impacted by deteriorating firm performance and/or directors' departures due to their unhappiness with governance or management. We define a dummy variable, *D(Terminate donations to charities affiliated with the director)*, to describe instances in which a firm stops making affiliated donations to a charity to which it donated in the previous year.

Panel A of Table 4 summarizes changes in donation propensities around director appointment and director departure. The results are consistent with our predictions. The probability of initiating affiliated donations is 12.9 percentage points higher in the year after the director's first appointment to the board

¹⁶ Alternatively, we could examine whether a firm starts donations when an independent director joins a charity board. However, the majority of independent directors' starting times at charities are missing. Private conversations with charity boards suggest that charity affiliations tend to last for many years, if not for a lifetime. The nature of the data does not allow us to test whether firms start donations when their independent directors join charity boards.

¹⁷ This measure of exogenous departure includes twenty-two cases in which the director died in the year of the departure or within two years of the departure. Only one out of these twenty-two directors' affiliated charities received corporate donations in the year of the director's departure.

Table 4
Changes in affiliated donations around director appointment and exogenous departure

Panel A: Univariate comparisons

i. Propensity of a firm to initiate donations to charities affiliated with a new independent director			
Became independent director the prior year?	N	D(Initiate donations to charity affiliated with the director)	
		Mean	Std. dev.
No	7,605	0.407	0.491
Yes	209	0.536	0.500
Difference (yes minus no)		0.129***	
t-statistic		3.67	
ii. Propensity of a firm to terminate donations to charities affiliated with a newly retired independent director			
Director retired the prior year?	N	D(Initiate donations to charity affiliated with the director)	
		Mean	Std. dev.
No	7,621	0.390	0.488
Yes	193	0.554	0.498
Difference (yes minus no)		0.165***	
t-statistic		4.53	
iii. Propensity of a firm to terminate donations to charities affiliated with a newly departed independent director who left all boards within two years			
Director departed the prior year and left all boards within two years?	N	D(Initiate donations to charity affiliated with the director)	
		Mean	Std. dev.
No	7,627	0.391	0.488
Yes	187	0.508	0.501
Difference (yes minus no)		0.117***	
t-statistic		3.15	
iv. Propensity of a firm to terminate donations to charities affiliated with an independent director who departed exogenously			
Director departed exogenously?	N	D(Initiate donations to charity affiliated with the director)	
		Mean	Std. dev.
No	7,538	0.389	0.488
Yes	276	0.514	0.501
Difference (yes minus no)		0.125***	
t-statistic		4.08	

(Continued)

that establishes the first connection between the firm and the charity than the baseline probability of 40.7% for other independent directors (part i of panel A). Conversely, the likelihood of terminating affiliated donations is 16.5 percentage points higher in the year after a director retires and severs the connection between the firm and the charity than the baseline probability of 39.0% for other independent directors (part ii of panel A). The likelihood of terminating affiliated donations is 11.7 percentage points higher in the year after a director suddenly departs than the baseline probability of 39.1% for other independent directors (part iii of panel A). Finally, the likelihood of terminating affiliated donations is 12.5 percentage points higher after

Table 4
(Continued)

Panel B: Regressions

Dependent variable	D(Initiate donations to charity affiliated with the director)	D(Terminate donations to charity affiliated with the director)		
	(1)	(2)	(3)	(4)
D(Became independent director the prior year)	0.131*** (3.60)			
D(Director retired the prior year)		0.122*** (2.81)		
D(Director departed the prior year and left all boards within two years)			0.098** (2.12)	
D(Director departed exogenously)				0.096** (2.34)
<i>Annual changes in:</i>				
D(CEO serving as Chairman)	0.071 (1.52)	0.054 (1.37)	0.053 (1.34)	0.054 (1.36)
CEO tenure	-0.004 (-0.75)	0.002 (0.34)	0.002 (0.35)	0.002 (0.35)
CEO ownership	-0.070 (-0.84)	-0.074 (-0.79)	-0.073 (-0.78)	-0.074 (-0.79)
Ln(No. of charities affiliated with non-co-opted independent directors)	0.003 (0.18)	0.002 (0.14)	0.003 (0.17)	0.002 (0.15)
Board size	-0.010 (-1.12)	0.009 (1.09)	0.009 (1.09)	0.009 (1.07)
D(Independent board)	-0.009 (-0.22)	-0.029 (-0.93)	-0.029 (-0.91)	-0.029 (-0.92)
D(Social connection(s) between independent director and CEO)	0.000 (0.00)	0.100** (2.32)	0.101** (2.33)	0.101** (2.33)
Top five institutions' ownership	0.030 (0.10)	0.042 (0.12)	0.036 (0.10)	0.035 (0.10)
Ln(Assets)	0.438*** (3.03)	0.111 (0.75)	0.111 (0.75)	0.112 (0.75)
Stock return	-0.021 (-0.47)	0.040 (1.20)	0.041 (1.22)	0.040 (1.19)
Profit margin	0.131 (0.68)	0.325* (1.75)	0.323* (1.74)	0.323* (1.75)
Market-to-book ratio	0.181*** (3.59)	0.014 (0.30)	0.013 (0.29)	0.015 (0.31)
Stock return volatility	0.066 (0.93)	-0.018 (-0.20)	-0.018 (-0.20)	-0.019 (-0.21)
Debt/assets	-0.126 (-0.33)	-0.121 (-0.37)	-0.123 (-0.38)	-0.117 (-0.35)

(Continued)

the exogenous departure of an independent director (aggregating retirement and sudden departure) than the baseline probability of 38.9% for other independent directors (part iv of panel A). These results are robust to the addition of controls for annual changes in governance and firm characteristics in multivariate regressions (panel B). Overall, corporate charitable donations tend to follow directors' moves, consistent with the notion that top executives use charitable contributions to influence independent directors' monitoring decisions.

Table 4
(Continued)

Dependent variable	D(Initiate donations to charity affiliated with the director) (1)	D(Terminate donations to charity affiliated with the director)		
		(2)	(3)	(4)
Advertisement/sales	0.342 (0.07)	12.324** (2.03)	12.298** (2.03)	12.297** (2.02)
R&D/assets	-0.989 (-0.49)	-0.409 (-0.16)	-0.425 (-0.17)	-0.456 (-0.18)
Industry growth	0.245** (2.48)	0.115 (1.18)	0.116 (1.19)	0.115 (1.17)
State corporate tax rate	-1.180 (-1.30)	-1.206 (-1.14)	-1.177 (-1.11)	-1.200 (-1.13)
Industry FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
No. of observations	7,814	7,814	7,814	7,814
Pseudo/Adjusted R^2	.145	.102	.102	.102

Tests in this table are conducted at the firm-director-year-charity level. Panel A reports univariate comparisons of a firm's propensity to initiate/terminate affiliated donations between subsamples of directors. The first block compares a firm's propensity to initiate donations to charities affiliated with a new independent director who establishes a connection between the firm and charities with its propensity of starting donations to charities affiliated with an incumbent director. The second block compares a firm's propensity to drop all of its donations to charities affiliated with an independent director who retired the prior year and cut the connection between the firm and charities with its propensity to drop donations to charities affiliated with an independent director who did not. Director retirement is defined as director departure from a firm at the age of seventy or older. The third block compares a firm's propensity to drop all of its donations to charities affiliated with an independent director who departed the board the prior year and left all boards within two years and cut the connection between the firm and charities with its propensity to drop donations to charities affiliated with an independent director who did not. An independent director departed exogenously if the director either retired the prior year or departed the prior year and left all boards within the last two years. The last block compares a firm's propensity to drop all of its donations to charities affiliated with an independent director whose exogenous departure cut the connection between the firm and charities with its propensity to drop donations to charities affiliated with an independent director who did not. Panel B reports the corresponding regression results. $D(\text{Initiate donations to charity affiliated with the director})$ equals one if a firm made donations to the charity affiliated with the director in the current year but not the prior year and zero otherwise. $D(\text{Terminate donations to charity affiliated with the director})$ equals one if a firm made donations to the charity affiliated with the director in the prior year but not the current year and zero otherwise. The control variables are annual changes of the corresponding variables, described in Appendix B. We report t -statistics based on heteroscedasticity-robust standard errors adjusted for firm clusters in parentheses below the corresponding regression coefficients. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

4. Effect of Affiliated Donations on CEO Compensation

4.1 Baseline specification

We examine the relationship between affiliated donations and the level of annual CEO compensation. We control for firm characteristics and governance measures, and we include industry fixed effects and year fixed effects in our regressions (Core et al. 1999; Faulkender and Yang 2010). The regression model is:

CEO pay

$$\begin{aligned} &= \alpha + \beta \cdot \text{Affiliated donation}_{it} + \gamma_1 \cdot D(\text{CEO serving as chairman})_{it} \\ &+ \gamma_2 \cdot \text{CEO tenure}_{it} + \gamma_3 \cdot \text{CEO ownership}_{it} + \gamma_4 \cdot \text{Board size}_{it} \end{aligned}$$

$$\begin{aligned}
& + \gamma_5 \cdot D(\text{Independent board})_{it} \\
& + \gamma_6 \cdot D(\text{Social connections(s) between independent directors and CEO})_{it} \\
& + \gamma_7 \cdot D(\text{Above median fraction of co-opted directors})_{it} \\
& + \gamma_8 \cdot \text{Top five institutions ownership}_{it} + \gamma_9 \cdot \ln(\text{Assets})_{it} \\
& + \gamma_{10} \cdot \text{Stock return}_{it} + \gamma_{11} \cdot \text{Stock return}_{it} + \gamma_{12} \cdot \text{ROA}_{it} + \gamma_{13} \cdot d\text{ROA}_{it-1} \\
& + \gamma_{14} \cdot (M/B)_{it} + \gamma_{15} \cdot \text{Stock return volatility}_{it} + \gamma_{16} \cdot (\text{Debt/assets})_{it} \\
& + \mu_j + v_t + \varepsilon_{it}.
\end{aligned} \tag{1}$$

We report the regression results in Table 5. The dependent variable is the logarithm of total annual compensation (*TDC1* in ExecuComp) in thousands of dollars in Columns 1–4 and *TDC1* in millions of dollars in Column 5. Because the levels of CEO pay and affiliated donations may have within-industry dependence, we use industry clustering instead of firm clustering in this and other applicable tables. Industry clustering also implicitly accounts for serial correlations.¹⁸ As Column 1 shows, the coefficient of *D(affiliated donations)* is 0.090 and is significant at the 1% level, suggesting that CEO annual compensation is on average 9.4% ($= \exp(0.090) - 1$) higher at firms that make affiliated donations than at firms that do not. Not surprisingly, we find that CEO compensation is greater when the CEO serves as the chairman of the board or has lower ownership, when the firm is larger, or when the firm has better stock return and lagged stock return, higher ROA, or a greater market-to-book value.

We next test the relationship between the magnitude of affiliated donations and the level of CEO compensation. We use four size measures for affiliated donations, including the logarithm of (1 + total dollar amount of affiliated donations), the total amount of affiliated donations scaled by average director pay, the total amount of affiliated donations scaled by the amount of all of the firm's charitable donations (affiliated and unaffiliated), and the total amount of affiliated donations expressed in millions of dollars. Due to missing or zero values of the scaling variables, the number of observations is smaller in Columns 3 and 4. CEO total pay and affiliated donation amount are winsorized at the 2.5th and the 97.5th percentiles to further reduce the impact of outliers in Column 5. The results confirm a significant positive correlation between the amount of affiliated donations (Columns 2–5) and CEO compensation. Column 5 reports that a \$1 million increase in affiliated donations is associated with an increase of \$424,000 in CEO pay. Our results are consistent with our hypothesis that

¹⁸ We thank an anonymous referee for suggesting clustering the standard errors by industry in the analyses on the level of CEO compensation. Clustering the standard errors by firm produces similar results.

affiliated donations could serve as a channel through which the CEO influences independent directors' compensation decisions.¹⁹

4.2 The compensation committee channel

To corroborate our interpretation of the above results that the association between affiliated donations and CEO pay is driven by a quid pro quo, we explore the variation in affiliated donations based on independent directors' committee assignment. We expect to find a strong correlation between affiliated donations and CEO compensation when the affiliated director serves on the compensation committee, which plays a vital role in determining CEO compensation. Among firms donating to charities affiliated with compensation committee members, we expect to find a stronger effect if the recipient charities are affiliated with the committee chair or a large fraction of compensation committee members.

We test these predictions by first splitting affiliated donations into two categories: donations to charities affiliated with any compensation committee member(s) and donations to charities affiliated with independent directors who do not serve on the compensation committee. Of the 917 firm-year observations with affiliated donations in our sample, 680 are affiliated with compensation committee members and 237 are not. The results reported in Table 6 are consistent with our hypothesis. CEOs earn 9.5% ($=\exp(0.091)-1$) more on average at firms that donate to charities affiliated with compensation committee members than at firms not making affiliated donations (Column 1). In contrast, the correlation between CEO compensation and affiliated donations unrelated to the compensation committee is statistically insignificant.

We next split the 680 observations with donations affiliated with the compensation committee into two groups along two dimensions. First, we split donations to charities affiliated with the committee chair (296 observations, of which fifty-seven involve the chair only) and those to charities related only to other members of the compensation committee (384 observations). On average, CEOs earn 15.6% ($=\exp(0.145)-1$) more at firms that donate to charities affiliated with the compensation committee chair than at firms not making affiliated donations (Column 2). In contrast, CEO pay is only 5.1% ($=\exp(0.050)-1$, statistically insignificant) higher at firms that donate to charities affiliated only with nonchair compensation committee members than at firms making no affiliated donations. The difference in coefficients between

¹⁹ The documented contemporaneous relationship between affiliated donations and CEO compensation could be consistent with two possibilities: (1) the CEO agrees to donate to independent directors' affiliated charities and subsequently receives a generous pay package (around the time when the donations are being recorded) and (2) CEO compensation decisions and corporate donation decisions are made around the same time. In a private conversation with a director serving on multiple charities, we confirmed that it takes time for charities to cultivate relationships. Donations may occur during a charity's capital campaign many months after the conversation between the charity and the donor. In either case, the positive relationship between affiliated donations and CEO compensation signals a quid pro quo relationship.

Table 5
Affiliated donations and CEO compensation

Dependent variable	Ln(Total pay)				Total pay
	(1)	(2)	(3)	(4)	(5)
D(Affiliated donations)	0.090*** (2.95)				
Ln(1+affiliated donations in thousands of dollars)		0.016*** (3.21)			
Ln(0.001+affiliated donation amount, scaled by average director pay)			0.011*** (3.25)		
Ln(0.001+affiliated donation amount, scaled by firm's total annual donations)				0.018*** (2.82)	
Affiliated donation amount (in millions of dollars)					0.424** (2.41)
D(CEO serving as chairman)	0.196*** (5.13)	0.195*** (5.10)	0.183*** (4.37)	0.109 (1.03)	1.151** (2.20)
CEO tenure	0.003 (0.58)	0.003 (0.57)	0.004 (0.69)	0.004 (0.47)	0.053 (1.51)
CEO ownership	-0.029* (-1.66)	-0.029 (-1.63)	-0.029* (-1.66)	-0.046 (-1.29)	-0.095 (-0.85)
Board size	0.016 (1.24)	0.016 (1.21)	0.014 (1.22)	-0.009 (-0.52)	0.057 (0.71)
D(Independent board)	0.049 (0.58)	0.051 (0.61)	0.066 (0.81)	0.274** (2.35)	0.003 (0.00)
D(Social connection(s) between independent directors and CEO)	-0.002 (-0.04)	-0.003 (-0.06)	0.000 (0.01)	0.057 (0.63)	0.014 (0.03)
D(Above-median fraction of co-opted directors)	-0.001 (-0.02)	-0.001 (-0.01)	0.001 (0.01)	0.008 (0.26)	0.131 (0.24)
Top five institutions' ownership	0.183 (0.96)	0.195 (1.02)	0.175 (0.90)	-0.310 (-1.21)	0.815 (0.52)
Ln(Assets)	0.350*** (7.93)	0.346*** (7.73)	0.351*** (8.27)	0.356*** (6.74)	3.261*** (11.83)
Stock return	0.170*** (3.69)	0.170*** (3.72)	0.171*** (4.90)	0.300*** (4.49)	1.365*** (4.63)
Lagged stock return	0.255*** (4.40)	0.256*** (4.39)	0.281*** (5.91)	0.318*** (3.74)	2.162*** (6.13)
ROA	0.963* (1.66)	0.971* (1.67)	0.860 (1.47)	0.959 (1.08)	5.778 (1.32)
Lagged dROA	0.307 (1.34)	0.313 (1.36)	0.292 (1.17)	0.247 (0.65)	2.039 (0.71)
Market-to-book ratio	0.055* (1.69)	0.053 (1.63)	0.062* (2.00)	0.074 (1.54)	0.757** (3.09)
Stock return volatility	0.024 (0.09)	0.025 (0.09)	0.097 (0.35)	-0.295 (-1.28)	0.185 (0.08)
Debt/assets	0.001 (0.01)	-0.003 (-0.03)	-0.029 (-0.25)	-0.077 (-0.51)	-0.605 (-0.32)
Industry FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes
No. of observations	3,746	3,746	3,700	1,398	3,746
Adjusted R^2	.312	.313	.315	.309	.336

This table examines the relationship between affiliated donations and CEO compensation. Total pay is annual direct compensation for the CEO (*TDC1* in ExecuComp) in thousands of dollars in Columns 1–4 and in millions of dollars in Column 5. All regressions include industry fixed effects and year fixed effects. Industries are Fama-French 12 industries. All variables are described in Appendix B. We report *t*-statistics based on heteroscedasticity-robust standard errors adjusted for industry clusters in parentheses below the corresponding regression coefficients. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

affiliated donations related to the compensation committee chair and those related to only nonchair members is statistically significant, with a *p*-value of .064.

Table 6
Affiliated donations and CEO compensation, roles of compensation committee

Dependent variable	Ln(Total pay)		
	(1)	(2)	(3)
D(Affiliated donations related to compensation committee)	0.091*** (3.07)		
D(Affiliated donations related to compensation committee chair)		0.145*** (4.57)	
D(Affiliated donations related to other compensation committee member)		0.050 (1.24)	
D(Affiliated donations related to above-median fraction of compensation committee members)			0.128*** (2.85)
D(Affiliated donations related to below-median fraction of compensation committee members)			0.055** (2.23)
D(Affiliated donations unrelated to compensation committee)	0.088 (1.51)	0.088 (1.52)	0.089 (1.55)
Controls (same as in Table 5)	Yes	Yes	Yes
Industry FEs	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
No. of observations	3,746	3,746	3,746
Adjusted R^2	.312	.313	.313
Test of difference (diff and [p -value])			
Comp committee vs. noncomp committee	0.003 [.963]		
Comp committee chair vs. member		0.095* [.064]	
Above-median vs. below-median fraction of comp committee members			0.073 [.154]

This table examines the role of compensation committees in the relationship between affiliated donations and CEO compensation. Total pay is annual direct compensation for the CEO (*TDC1* in ExecuComp) in thousands of dollars. All regressions include industry fixed effects and year fixed effects. Industries are Fama-French 12 industries. All regressions include the same firm and CEO control variables as in Table 5. All variables are described in Appendix B. We report t -statistics based on heteroscedasticity-robust standard errors adjusted for industry clusters in parentheses below the corresponding regression coefficients. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The bottom of the table reports the results (coefficient differences and p -values) from statistical tests of the difference in the coefficients on *D(Affiliated donations related to compensation committee)* and *D(Affiliated donations unrelated to compensation committee)* in Column 1, the difference in the coefficients on *D(Affiliated donations related to compensation committee chair)* and *D(Affiliated donations related to other compensation committee member)* in Column 2, and the difference in the coefficients on *D(Affiliated donations related to above-median fraction of compensation committee members)* and *D(Affiliated donations related to below-median fraction of compensation committee members)* in Column 3.

Second, we compare firms that make affiliated donations to an above-median fraction of compensation committee members (356 observations) with firms that make affiliated donations to a below-median fraction of committee members (324 observations). On average, CEOs earn 13.7% ($=\exp(0.128) - 1$) more at firms that donate to charities affiliated with an above-median fraction of compensation committee members than at firms that do not make affiliated donations.²⁰ The difference in coefficients between affiliated donations related to an above-median fraction of compensation committee members and those

²⁰ The median fraction of the compensation committee members corresponds to 1.7 compensation committee members in our sample. For robustness, we also separate affiliated donations related to multiple compensation committee members from those related to only one committee member. We obtain similar results using this construct.

related to a below-median fraction of committee members is statistically insignificant at a conventional level, with a p -value of .154.

In all specifications, we include an indicator for the 237 affiliated donations unrelated to the compensation committee. As expected, we do not find significant links between CEO compensation and this variable. Overall, our cross-sectional tests exploiting affiliated directors' roles on the compensation committee are consistent with the interpretation that affiliated donations could serve as a channel through which the CEO gains leverage over independent directors in exchange for a more generous compensation package.

5. Robustness Tests and Alternative Explanations

In this section, we conduct analyses to alleviate potential endogeneity concerns and rule out alternative hypotheses. Specifically, some firm, CEO, or director characteristics omitted from our regression specifications (e.g., free cash flow, connections between CEOs and independent directors via the charities receiving affiliated donations) could affect both affiliated donation and CEO compensation decisions and drive the positive relationship between the two. Another potential concern is that firms making affiliated donations could differ systematically from firms not making such donations.

5.1 Changes in CEO compensation when firms commence and terminate affiliated donations

To mitigate the concern that firms making affiliated donations may differ from other firms on dimensions not accounted for in our regression analyses, we examine the change in CEO compensation when a firm commences (or terminates) affiliated donations. To do so, we regress the annual change in CEO compensation on a dummy variable that equals one in the year when a firm makes an affiliated donation for the first time (within our sample) and zero otherwise. We expect a larger increase in CEO compensation when firms begin affiliated donations. Because examining annual changes parses out firm-specific and time-invariant components of CEO compensation, such evidence would further support our interpretation of a genuine association between affiliated donations and CEO compensation.

In a similar vein, we examine the change in CEO compensation when a firm terminates all affiliated donations. We construct a dummy variable that equals one in the year when a firm ceases making affiliated donations and zero otherwise. We regress the change in CEO compensation on the termination dummy variable and expect to find a negative correlation. We further contrast the initiation or termination of affiliated donations involving directors serving on the compensation committee with the initiation or termination of all other affiliated donations. In the regressions, we control for common determinants of CEO compensation, including corporate governance and firm characteristics.

Table 7
Changes in CEO compensation around initiation and termination of affiliated donations

Panel A: Summary statistics

Variable	<i>N</i>	Mean	Std. dev.
Annual change in total CEO pay (in millions of dollars)	3,403	0.380	5.490
D(Initiation of affiliated donations)	3,403	0.025	0.157
D(Initiation of affiliated donations related to compensation committee)	3,403	0.015	0.120
D(Initiation of affiliated donations unrelated to compensation committee)	3,403	0.011	0.102
D(Termination of affiliated donations)	3,403	0.032	0.177
D(Termination of affiliated donations related to compensation committee)	3,403	0.020	0.140
D(Termination of affiliated donations unrelated to compensation committee)	3,403	0.012	0.110

(Continued)

We also include annual changes in sales and net income as additional controls for firm performance.

Panel A of Table 7 reports the summary statistics of the main variables. The average annual change in CEO pay is \$380,000. In 2.5% of the firm-year observations, affiliated donations are initiated; in 3.2% of the observations, they are terminated. Panel B reports the regression results. When a firm initiates affiliated donations, the increase in CEO pay is \$1.3 million higher than in other years (Column 1). This is driven by cases in which the firm initiates affiliated donations involving its compensation committee members (Column 2). When a firm terminates all of its affiliated donations, the change in CEO pay is \$1.1 million lower than in other years (Column 3); this result is also driven by cases in which the firm terminates affiliated donations involving its compensation committee members (Column 4).

5.2 Fixed effects, propensity score matching, and alternative explanations

We further address endogeneity concerns using three fixed-effect specifications. In the first specification, we replace industry fixed effects with firm fixed effects, which parse out firm-specific and time-invariant omitted factors that affect both affiliated donations and CEO compensation. The second specification includes firm and CEO interactive fixed effects that control for time-invariant firm and CEO characteristics (e.g., a CEO’s generosity or monetary preference) that may drive the variations in affiliated donations and CEO compensation. We report in Columns 1 and 2 of Table 8 that the positive association between affiliated donations and CEO compensation is significant at the 10% and 5% levels, respectively. The coefficient estimate for affiliated donations declines from 0.090 in the baseline specification (Column 1 of Table 5) to 0.043 and 0.062, respectively, but remains economically meaningful. In these specifications, fixed effects absorb cross-sectional variations, and the estimated effect comes solely from the time series variations in the data. Given that we have only ten

Table 7
(Continued)

Panel B: Regressions

Dependent variable	Annual change in total pay			
	(1)	(2)	(3)	(4)
D(Initiation of affiliated donations)	1.318*** (2.87)			
D(Initiation of affiliated donations related to compensation committee)		1.753*** (2.68)		
D(Initiation of affiliated donations unrelated to compensation committee)		0.717 (1.21)		
D(Termination of affiliated donations)			-1.148* (-1.89)	
D(Termination of affiliated donations related to compensation committee)				-1.325* (-1.82)
D(Termination of affiliated donations unrelated to compensation committee)				-0.862 (-0.84)
D(CEO serving as chairman)	0.101 (0.40)	0.084 (0.33)	0.114 (0.45)	0.113 (0.45)
CEO tenure	0.014 (0.82)	0.014 (0.80)	0.013 (0.79)	0.013 (0.79)
CEO ownership	0.008 (0.25)	0.008 (0.24)	0.008 (0.27)	0.008 (0.28)
Board size	0.035 (1.00)	0.036 (1.04)	0.041 (1.18)	0.040 (1.17)
D(Independent board)	-0.400* (-1.73)	-0.395* (-1.71)	-0.338 (-1.47)	-0.339 (-1.48)
D(Social connection(s) between independent directors and CEO)	0.015 (0.11)	0.015 (0.11)	0.027 (0.20)	0.030 (0.22)
D(Above-median fraction of co-opted directors)	-0.022 (-0.13)	-0.021 (-0.13)	-0.016 (-0.10)	-0.018 (-0.10)
Top five institutions' ownership	2.425*** (2.60)	2.406** (2.57)	2.353*** (2.53)	2.351** (2.52)
Ln(Assets)	0.050 (0.60)	0.047 (0.57)	0.034 (0.41)	0.034 (0.40)
Stock return	1.941*** (4.55)	1.942*** (4.55)	1.929*** (4.52)	1.927*** (4.51)
Lagged stock return	1.623*** (4.20)	1.616*** (4.16)	1.589*** (4.10)	1.588*** (4.09)
ROA	-0.012 (-0.01)	0.001 (0.00)	0.138 (0.10)	0.123 (0.09)
Lagged dROA	-1.011 (-0.35)	-0.987 (-0.34)	-1.085 (-0.37)	-1.066 (-0.37)
Market-to-book ratio	-0.100 (-0.96)	-0.099 (-0.95)	-0.110 (-1.07)	-0.110 (-1.06)
Stock return volatility	-1.459** (-2.10)	-1.465** (-2.10)	-1.425** (-2.05)	-1.431** (-2.06)
Debt/assets	0.203 (0.57)	0.207 (0.58)	0.191 (0.53)	0.194 (0.54)
Annual change in sales in billions of dollars	0.024 (0.56)	0.024 (0.55)	0.025 (0.58)	0.025 (0.58)
Annual change in net income in billions of dollars	0.380*** (2.83)	0.380*** (2.83)	0.371*** (2.76)	0.371*** (2.76)

(Continued)

years of data and that the affiliated donation dummy does not change over time for approximately two-thirds of the firms, the estimated effect is likely partial. Thus, we do not include firm fixed effects in subsequent tests.

The third specification examines the possibility that an indulgent director, who is likely to award a generous pay package to the CEO, also solicits the

Table 7
(Continued)

Dependent variable	Annual change in total pay			
	(1)	(2)	(3)	(4)
Industry FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
No. of observations	3,403	3,403	3,403	3,403
Adjusted R^2	.039	.038	.039	.039

This table examines CEO pay changes around initiation and termination of affiliated donations. Panel A summarizes the main variables, and panel B reports the regression results. The dependent variable is the annual change in total CEO pay in millions of dollars. For each year, $D(\text{Initiation of affiliated donations})$ equals one if a firm makes its first affiliated donation that year and zero otherwise. $D(\text{Initiation of affiliated donations related to compensation committee})$ equals one if a firm makes its first affiliated donation that year and the donation goes to a charity affiliated with an independent director on the compensation committee and zero otherwise. $D(\text{Initiation of affiliated donations unrelated to compensation committee})$ equals one if a firm makes its first affiliated donation that year and the donation goes to a charity affiliated with an independent director outside the compensation committee and zero otherwise. $D(\text{Termination of affiliated donations})$ equals one if a firm stops donating to all charities that are affiliated with independent directors and zero otherwise. $D(\text{Termination of affiliated donations related to compensation committee})$ equals one if a firm stops all affiliated donations and the last donation goes to a charity affiliated with independent director(s) on the compensation committee and zero otherwise. $D(\text{Termination of affiliated donations unrelated to compensation committee})$ equals one if a firm stops all affiliated donations and the last donation goes to a charity affiliated with independent director(s) outside the compensation committee and zero otherwise. All regressions include industry fixed effects and year fixed effects. Industries are Fama-French 12 industries. All variables are described in Appendix B. We report t -statistics based on heteroscedasticity-robust standard errors adjusted for firm clusters in parentheses below the corresponding regression coefficients. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

firm’s donations to his or her affiliated charities, driving the positive relationship between the two factors. To alleviate such concerns, we conduct an analysis that includes director fixed effects. Specifically, we form a subsample of firms that have at least one independent director serving on the corporate boards of two S&P 500 firms in the same year, only one of which donates to charities affiliated with the director. We then run the baseline regression using these 1,926 director-year observations and include director fixed effects in addition to industry fixed effects and year fixed effects. Thus, the regression specification parses out time-invariant director characteristics that affect both affiliated donations and CEO compensation. The remaining variation captured by the coefficient estimate of affiliated donations for a given director results from that director’s charities receiving (or not receiving) corporate donations. The results presented in Column 3 of Table 8 are consistent with the baseline results reported in Table 5. On average, CEOs at firms that make affiliated donations earn 8.2% ($=\exp(0.079)-1$) more than their counterparts at other firms.

To further investigate potential systematic differences between firms that make affiliated donations and firms that do not, we use the propensity score matching (PSM) approach to control for observable differences in firm and CEO characteristics. The matching model is a logit model that estimates the propensity to make affiliated donations based on the explanatory variables in panel A of Table 3, including year fixed effects and industry fixed effects. Using a caliper equal to 0.25 standard deviations of the estimated propensity score without replacement, we are able to find close matches for 751 observations of

Table 8
Affiliated donations and CEO compensation, fixed effects, PSM, and alternative explanations

	Ln(Total pay)								
	Fixed effects			PSM	Alternative explanations				
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
D(Affiliated donations)	0.043* (1.73)	0.062** (2.07)	0.079** (2.40)	0.119*** (2.70)	0.105*** (2.75)	0.084*** (3.57)	0.079*** (2.83)	0.091** (2.50)	0.100*** (2.65)
D(Unaffiliated donations)					0.043 (1.05)				
D(CEO-affiliated donations)						0.019 (0.55)			
D(Inside director-affiliated donations)							0.025 (0.57)		
D(CEO is connected to independent director(s) via affiliated charity)								0.144 (1.09)	
Free cash flow									0.520 (0.77)
Controls (same as Table 5)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other FEs	Firm	Firm	Industry, director	Industry	Industry	Industry	Industry	Industry	Industry
No. of observations	3,746	3,746	1,926	1,502	3,746	3,746	3,746	3,746	3,449
Adjusted R ²	.636	.685	.470	.316	.311	.311	.311	.311	.336

This table examines the relationship between affiliated donations and CEO compensation in various alternative specifications. The dependent variable is the logarithm of total CEO pay in thousands of dollars. Column 1 includes firm fixed effects. Column 2 includes firm-by-CEO fixed effects. In Column 3, the sample includes directors who serve on the boards of two S&P 500 firms in a given year, where *D(Affiliated donations)* equals one for only one of the two firms. Director fixed effects are included. In Column 4, the sample includes firms making affiliated donations and matching firms not making affiliated donations. The propensity score matching model is estimated with a caliper of 0.05 (i.e., 0.25 of the standard deviation of the estimated propensity score). The matching model is a logit model that estimates the propensity to make affiliated donations based on the explanatory variables in Table 3, panel A. Column 5 controls for unaffiliated donations. Column 6 controls for CEO-affiliated donations. Column 7 controls for donations to charities affiliated with all inside directors (CEOs or non-CEO inside directors). Column 8 controls for social connections between the CEO and independent director(s) via the affiliated charities receiving donations. Column 9 controls for free cash flow. Industries are Fama-French 12 industries. All variables are described in Appendix B. We report *t*-statistics based on heteroscedasticity-robust standard errors adjusted for industry clusters in parentheses below the corresponding regression coefficients. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

affiliated donations. There is no significant difference for any of the matching variables within a pair of matched observations (untabulated). We show that CEO pay is 12.6% ($=\exp(0.119)-1$) higher at firms that make affiliated donations than at firms that do not make affiliated donations (Column 4).

In Columns 5–9, we test alternative hypotheses. To show that the effect we document is not present among all corporate donations, we control for whether firms make unaffiliated donations and report the results in Column 5. The coefficient of *D(Affiliated donations)* remains positive and significant, while the coefficient of *D(Unaffiliated donations)* is not significant, suggesting that donations affiliated with independent directors have a real impact on CEO compensation.

Masulis and Reza (2015) suggest that corporate giving may reflect CEOs' charitable interests. To address this concern, we contrast the effect of independent director–affiliated donations on CEO compensation with that of CEO-affiliated donations. We show in Column 6 that independent director–affiliated donations are significant determinants of CEO compensation (a coefficient estimate of 0.084, significant at the 1% level), but CEO-affiliated donations are not (a coefficient estimate of 0.019, statistically insignificant). Thus, the effect of affiliated donations on CEO compensation is not merely a manifestation of the effect of CEO-affiliated donations documented in Masulis and Reza (2015). In a similar vein, we next investigate whether our results are driven by donations to charities affiliated with inside directors (including CEOs and non-CEO inside directors). We include a control variable for inside director–affiliated donations. As Column 7 shows, independent director–affiliated donations remain a significant determinant of CEO pay, while inside director–affiliated donations are not related to CEO pay.

Another possibility is that the social ties between CEOs and independent directors developed at charities drive both the affiliated donation and CEO compensation decisions.²¹ To address such concerns, we include a control variable for the social ties that any independent directors have with the CEO via the affiliated charities. We show in Column 8 that the association between affiliated donations and CEO pay continues to be economically and statistically significant. Our results, reported in Columns 6–8, suggest that the effect of affiliated donations on CEO compensation is not a reflection of common charitable interests or social ties between the CEO (or other inside directors) and independent directors.

One may be concerned that other governance issues such as free cash flow may lead to affiliated donations and higher CEO pay, which is a different type of agency problem. To address this concern, we control for free cash flow and report the results in Column 9. We calculate free cash flow as operating

²¹ Among charities that receive affiliated donations through independent directors, 11.5% are affiliated with the CEO of the donating firm, 9.8% of whom are the chair of the board (COB). Non-CEO COBs collectively add only 1.9% of ties to recipient charities, through which they may develop ties with the affiliated directors.

income before depreciation minus interest expenses, income taxes, and capital expenditures, scaled by book value of total assets (Masulis, Wang, and Xie 2007). We leave ROA out of the model to avoid multicollinearity. We find that free cash flow has no explanatory power for CEO total pay. More importantly, the relationship between affiliated donations and CEO compensation remains economically and statistically significant.²²

Overall, these additional analyses suggest that the positive link between affiliated donations and CEO compensation likely captures a quid pro quo relationship between the CEO and independent directors, rather than reflecting firms' selection into making affiliated donations.

5.3 Instrumental variable specification

It is impossible to exhaust all potential omitted variables that may drive the positive correlation between affiliated donations and CEO compensation. To further address endogeneity concerns, we construct an instrumental variable for making affiliated donations at the firm.²³ More specifically, we identify an exogenous source of variation in the demand for affiliated donations and study whether CEO compensation increases at firms making donations in response to such a demand shock. Focusing on universities affiliated with independent corporate directors, we identify forced turnovers of their basketball and football head coaches. Forced turnovers of these college head coaches occur frequently. They can also be very expensive: replacing a head coach for poor performance can cost a university millions of dollars. The costs consist of (1) an increase in the compensation package for the incoming coach (relative to that of the outgoing coach), (2) severance pay for the outgoing coach, (3) a buyout for the incoming coach if there is a contractual obligation to the previous employer, (4) funds to replace assistant coaches associated with the turnover, and in some cases (5) associated legal expenses. We relate one recent example that details the expenses of replacing a college football coach in Appendix A.

Charitable contributions and rights/licensing are the two main sources of athletic department revenue at many universities. Before a forced coach turnover, an athletic team has typically performed poorly for some time, which decreases both rights/licensing revenue and contributions. Thus, replacing a head coach substantially increases the pressure to fundraise. At these

²² In unreported regressions, we use two alternative measures of free cash flow: operating income before depreciation minus interest expenses, income taxes, and dividends, scaled by book value of total assets (Lang, Stulz, and Walkling 1991), and operating income before depreciation minus interest expenses, income taxes, dividends, and capital expenditures, scaled by book value of total assets. Neither measure is significantly correlated with CEO compensation; the positive link between affiliated donations and CEO compensation remains significant.

²³ Ex ante, our ordinary least squares (OLS) estimate may either overestimate or underestimate the population average treatment effect (ATE). For example, firms with free cash flows may make more affiliated donations and pay their CEOs more, and such omitted variables would lead the OLS estimate to overstate the true effect (affirmative endogeneity, Jiang 2017). Alternatively, CEOs with lower compensation may be more motivated to allocate company donations to independent directors' affiliated charities to build the quid pro quo relationship, causing the OLS estimate to understate the true effect (corrective endogeneity, Jiang 2017).

Table 9
Affiliated donations and CEO compensation, instrumental variable (IV) specification

Panel A: Summary statistics

Year	No. of firms	No. of firms with multiple directors' affiliated universities firing head coach	No. of unique universities firing head coach	
			Basketball	Football
2003	311	79	16	17
2004	316	120	18	17
2005	327	100	25	9
2006	356	187	34	12
2007	374	120	21	13
2008	382	121	15	21
2009	406	139	28	14
2010	415	152	25	20
2011	427	155	21	24
2012	432	148	15	13
Total	3,746	1,321	218	160

(Continued)

critical moments, independent directors drum up contributions to universities and greatly appreciate CEOs' understanding and financial support, which may weaken directors' monitoring incentive. This helps with the relevance restriction of the instrumental variable. Unlike a shock to the supply of affiliated donations (on the corporate side), a shock to the demand (on the charity side) is unlikely to be correlated with omitted factors at the donating firm that affect CEO compensation. This helps with the exclusion restriction of the instrumental variable.

We compile and cross-check information on college basketball and football coach turnovers from sportingnews.com, cbssports.com, collegesportsinfo.com, coachesbythenumbers.com, and Wikipedia. We classify a coach turnover as forced (1) if the announcement explicitly states that the coach was "fired," (2) if the coach "retired" with poor performance and not for health reasons, and (3) if the coach "resigned" with poor performance and without taking a comparable or better position immediately. We identify 218 forced basketball coach turnovers and 160 forced football coach turnovers during our sample period between 2003 and 2012. We run instrumental variable regression tests at the firm-year level involving all independent directors affiliated with universities. The instrumental variable, $D(\text{Forced coach turnovers})$, equals one if multiple independent directors' affiliated universities fire coaches and zero otherwise.²⁴ Panel A of Table 9 shows that of the 3,746 firm-year observations, 1,321 observations involve forced coach turnovers.

Regarding the timing of fundraising for coach firing and affiliated donations in response to coach firing, we find that forced coach turnovers are typically

²⁴ We also consider defining the instrumental variable based on at least one independent director or three or more independent directors affiliated with universities firing coaches. The instrument based on at least one independent director is a weak instrument in the first stage, with an F -statistic of 4.4. The instrument based on three or more independent directors is a strong instrument, and our results are robust to using that instrument.

Table 9
(Continued)

Panel B: Regressions

Dependent variable	First-stage estimation D(Affiliated donations) (1)	Second-stage estimation Ln(Total pay) (2)
D(Forced coach turnovers)	0.080*** (4.86)	
D(Affiliated donations), predicted value from (1)		0.665* (1.72)
D(CEO serving as chairman)	0.066** (2.39)	0.152*** (2.65)
CEO tenure	0.001 (0.69)	0.004 (0.64)
CEO ownership	-0.015*** (-4.42)	-0.025 (-1.50)
Board size	0.031*** (4.85)	-0.003 (-0.15)
D(Independent board)	0.037 (0.93)	0.026 (0.26)
D(Social connection(s) between independent directors and CEO)	0.042* (1.70)	-0.026 (-0.57)
D(Above-median fraction of co-opted directors)	-0.067*** (-2.88)	0.034 (0.77)
Top five institutions' ownership	-0.200 (-1.35)	0.362 (1.33)
Ln(Assets)	0.092*** (5.18)	0.302*** (5.22)
Stock return	-0.021 (-0.80)	0.204*** (3.21)
Lagged stock return	-0.024 (-0.97)	0.304*** (4.70)
ROA	0.282 (1.42)	0.843* (1.81)
Lagged dROA	0.092 (0.74)	0.292 (1.04)
Market-to-book ratio	0.023* (1.79)	0.047 (1.36)
Stock return volatility	0.082 (0.92)	0.063 (0.44)
Debt/assets	0.031 (0.38)	-0.018 (-0.13)
Industry FEs	Yes	Yes
Year FEs	Yes	Yes
No. of observations	3,746	3,746
Adjusted R^2	.195	.239
F-statistic (coefficient estimate for IV = 0)	23.62	

This table reports the results from the IV regression determining the effect of affiliated donations on CEO compensation. Panel A describes the distribution of forced head coach turnovers by year. Panel B reports regression results. The IV, $D(\text{Forced coach turnovers})$, equals one if multiple independent directors' affiliated universities fire head basketball or football coaches and zero otherwise. Column 1 summarizes the first-stage result based on the ordinary least squares regression model. The dependent variable is $D(\text{Affiliated donations})$, which equals one if a company donates to a university with which its independent director is affiliated and zero otherwise. The predicted value for $D(\text{Affiliated donations})$ from Column 1 is then used in the second-stage regression, which is reported in Column 2. The dependent variable is the logarithm of total CEO pay in thousands of dollars. All regressions include industry fixed effects and year fixed effects. Industries are Fama-French 12 industries. All variables are described in Appendix B. We report t -statistics based on heteroscedasticity-robust standard errors adjusted for firm clusters in parentheses below the corresponding regression coefficients. The F -statistic in Column 1 is for the test of the null hypothesis that the coefficient estimate for the IV is zero. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

deliberated over a period of time, and termination for poor performance is rarely unexpected. Therefore, colleges may ramp up fundraising activities many months before actually firing their head coaches. Thus, we use the forced coach turnover dummy as the instrumental variable for the affiliated donation dummy, and we re-estimate the CEO total pay regression model and report the results in panel B of Table 9. The first-stage result in Column 1 shows a strong correlation between affiliated donations and college football/basketball forced coach turnovers. The F -statistic is 23.62 for the test of the null hypothesis that the coefficient estimate for the instrumental variable equals zero, exceeding the cutoff for the weak instrument test.²⁵ More importantly, the second-stage result reported in Column 2 reinforces the positive link between affiliated donations and CEO compensation; that is, CEO pay increases as affiliated donations increase for an exogenous reason. This result is consistent with our interpretation that affiliated donations affect CEO compensation in a causal manner.²⁶

6. CEO Replacement Decisions and Firm Performance

Analyses reported in Sections 4 and 5 focus on the effect of affiliated donations on the level of CEO compensation. This relationship is significant primarily when a firm donates to charities affiliated with its compensation committee members. In this section, we examine how affiliated donations affect another aspect of boards' monitoring decisions: the sensitivity of forced CEO turnover to firm performance. We further test the effect of affiliated donations on firm performance.

6.1 Affiliated donations and the sensitivity of forced CEO turnover to performance

If CEOs use affiliated donations to cultivate relationships with independent directors, we expect the practice to lower the sensitivity of forced CEO turnover

²⁵ We also consider the effect of unforced coach turnovers in a placebo test. Unforced coach turnovers can still impose significant costs on a university, but the costs are not as substantial as those for forced coach turnovers (e.g., severance pay is not needed for unforced coach turnovers). We replace forced coach turnovers with unforced coach turnovers in the first-stage regression and find that affiliated donations are not significantly associated with unforced coach turnovers. We do not report the result for brevity.

²⁶ Our IV estimator is larger than the OLS estimator, consistent with survey findings in Card (1999) and Jiang (2017). One possible explanation is that the IV estimate reflects the local average treatment effect (LATE), and the OLS estimate captures the population ATE. In particular, the IV estimator reflects the effect of affiliated donations on CEO pay when firms make donations in response to the increased demand for funds resulting from a university coach's firing. A director's willingness to engage in a quid pro quo could be much stronger when his or her affiliated university receives corporate donations when it desperately needs the funds. Thus, we caution that the LATE should not be considered the ATE, because the former is based on a slice of "compliers"—firms donating to affiliated universities that fired their coaches. Acknowledging that our IV approach cannot address endogeneity perfectly, we complement it by analyzing the compensation committee channel and changes in compensation around the initiation and termination of affiliated donations, adding various fixed effects, and using PSM. All these analyses provide consistent evidence supporting the causal impact of affiliated donations on CEO compensation.

to firm performance. We use Parrino's (1997) definition of forced turnover.²⁷ The probability of forced CEO turnover is 3.2% among firm-years with affiliated donations and 2.3% among firm-years without affiliated donations. The difference is not statistically significant at a conventional level (p -value = .144). We regress forced CEO turnover on contemporaneous stock return, lagged stock return, and $\text{Ln}(\text{Sales})$ following Kaplan and Minton (2012) and Gao et al. (2012, 2017). Following Jenter and Kanaan (2015), we add dummy variables indicating whether a CEO's age is between 63 and 66 (retirement age) and whether a CEO owns more than the sample median fraction of his or her company's stock. While forced CEO turnover is negatively correlated with stock returns, we predict that this correlation will be weaker at firms making affiliated donations.

We first contrast firms that make affiliated donations to a larger fraction of independent directors (a fraction greater than the sample median) with the remaining firm years. We then contrast firms that make larger affiliated donations (above the sample median value) with those not making such donations. Table 10 summarizes the regression results.

Absent substantial affiliated donations, CEOs are more likely to be replaced when lagged and contemporaneous stock returns are poor (Columns 1 and 3). In contrast, when a firm donates intensively to charities affiliated with independent directors, forced CEO turnover is not sensitive to stock returns (Columns 2 and 4). The difference in the coefficient estimates of the contemporaneous stock return between firms with and without intensive affiliated donations is statistically significant (p -values of .013 and .006 for the differences between Columns 1 and 2 and between Columns 3 and 4, respectively). The difference in the turnover-to-lagged-performance sensitivity between the two groups is statistically insignificant.²⁸ Overall, these results suggest that making large donations or donating to a critical mass of independent directors helps poorly performing CEOs retain their jobs.

We also examine the role of the governance committee in replacing poorly performing CEOs. Donations made to charities affiliated with governance committee members do not appear to attenuate CEO turnover incrementally,

²⁷ A CEO turnover is classified as forced if the press reports that the CEO was fired or forced out, or retired or resigned, due to pressure. All other departures for CEOs age sixty and above are classified as voluntary. Departures for CEOs under age sixty are classified as forced if either (1) the press does not report the reason as death, poor health, or the acceptance of another position (including the chairmanship of the board) or (2) the press reports that the CEO is retiring but does not announce the retirement at least six months before the turnover.

²⁸ In alternative specifications, we use the whole sample and include one of the two indicators for intensive affiliated donations, as well as its interaction with stock return and its interaction with lagged stock return, in addition to its interaction with all control variables. In both specifications, we find negative loadings on the intensive affiliated donation indicator, stock return, and lagged stock return. In addition, we find a positive loading on the interaction of the intensive affiliated donation indicator and stock return. These findings are consistent with the results reported in Table 10. We also reach the same conclusion using a hazard model (instead of a probit model) in the forced CEO turnover regression, as in Jenter and Kanaan (2015). Moreover, if we replace stock returns with industry median-adjusted stock returns, or if we replace industry fixed effects and year fixed effects with industry-by-year fixed effects, we obtain similar results.

Table 10
Affiliated donations and the sensitivity of forced CEO turnover to performance

Dependent variable	D(Forced CEO turnover)			
	Above-median fraction of independent directors receiving affiliated donations		Above-median affiliated donation amount	
	No (1)	Yes (2)	No (3)	Yes (4)
Stock return	−0.027*** (−3.31)	0.055 (1.29)	−0.028*** (−3.45)	0.053 (1.64)
Lagged stock return	−0.036*** (−4.41)	−0.014 (−0.36)	−0.035*** (−4.34)	−0.023 (−0.86)
Ln(Sales)	0.002 (1.37)	0.021 (1.55)	0.003 (1.61)	0.019** (2.01)
D(CEO age between 63 and 66)	0.012 (1.11)	0.058 (0.84)	0.003 (0.35)	0.101 (1.45)
D(CEO ownership is above median)	−0.021*** (−4.10)	0.028 (0.93)	−0.018*** (−3.55)	−0.006 (−0.42)
Industry FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
No. of observations	3,625	492	3,633	484
Pseudo R^2	.106	.095	.101	.189
<i>p</i> -value (Col1 = Col2 or Col3 = Col4) on:				
Stock return		.013		.006
Lagged stock return		.229		.464

This table examines the relationship between affiliated donations and the sensitivity of forced CEO turnover to performance. The dependent variable is forced CEO turnover, which is equal to one if there is a forced CEO turnover in the current year and zero if there is no CEO turnover. Probit models are used, and marginal effects of the coefficients are reported. The subsamples are formed based on two different measures of affiliated donation intensity. All regressions include industry fixed effects and year fixed effects. Industries are Fama-French 12 industries. All variables are described in Appendix B. We report *t*-statistics based on heteroscedasticity-robust standard errors adjusted for firm clusters in parentheses below the corresponding regression coefficients. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

and this result is not surprising. “The governance committee might be assigned to fact finding—perhaps along with the audit committee, if a financial defalcation is the cause—but these committees are not empowered to fire CEOs,” observes Richard J. Mahoney, a former CEO of Monsanto and former director of Union Pacific Railroad and MetLife. Instead, he says, “The firing of a CEO would definitely require extended deliberations with the full board over a period of time.” Thus, targeted affiliated donations are unlikely to offer blanket insurance for a poorly performing CEO.

6.2 Affiliated donations and firm performance

If affiliated donations reflect a quid pro quo between independent directors and the CEO at shareholders’ expense, firm performance should suffer. We examine both the stock performance and accounting performance of firms making affiliated donations. The tests follow the regression specifications in Core et al. (1999) and Faulkender and Yang (2013). We cluster the standard errors at the industry level when examining stock performance because stock returns can be correlated within industries. Results are similar if we cluster the

Table 11
Affiliated donations and firm performance

Panel A: Stock performance		
Dependent variable	Stock return (1)	Tobin's Q (2)
D(Affiliated donations)	-0.024*** (-2.61)	-0.239*** (-2.65)
Stock return volatility	0.175* (1.70)	1.386** (2.27)
Ln(Market equity)	0.005 (0.66)	0.169*** (3.24)
ROA	-0.426*** (-4.23)	9.028*** (9.66)
Tobin's Q	0.081*** (9.02)	
Industry FEs	Yes	Yes
Year FEs	Yes	Yes
No. of observations	4,350	4,351
Adjusted R ²	.439	.531
Panel B: Accounting performance		
Dependent variable	ROA (1)	Sales growth (2)
D(Affiliated donations)	0.002 (0.31)	-0.012** (-1.98)
Industry cash flow volatility	-0.030 (-0.94)	-0.024 (-0.37)
Ln(Sales)	-0.006** (-2.23)	-0.008** (-2.54)
ROA		0.362*** (6.52)
Industry FEs	Yes	Yes
Year FEs	Yes	Yes
No. of observations	4,356	4,355
Adjusted R ²	.314	.190

This table examines the relationship between affiliated donations and firm performance. Panel A examines stock performance, including annual stock return and Tobin's Q. Panel B examines accounting performance, including ROA and sales growth. All regressions include industry fixed effects and year fixed effects. Industries are Fama-French 12 industries. All variables are described in Appendix B. We report *t*-statistics based on heteroscedasticity-robust standard errors adjusted for industry clusters (panel A) or firm clusters (panel B) in parentheses below the corresponding regression coefficients. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

standard errors by firm. For the tests of accounting performance, we cluster the standard errors by firm.

Table 11 summarizes the regression results. Controlling for stock return volatility, Ln(Market equity), ROA, Tobin's Q, industry fixed effects, and year fixed effects, we find that both annual stock returns and firm value (measured by Tobin's Q) are significantly lower when the firm makes affiliated donations. The differences are economically nontrivial. Stock returns are 2.4 percentage points lower and Tobin's Q is 0.239 lower for firms making affiliated donations than for firms not making such donations. Sales growth is also significantly lower for firms making affiliated donations (by 1.2 percentage points) after we control for industry cash flow volatility, Ln(sales), ROA, industry fixed effects,

and year fixed effects.²⁹ ROA, however, is not significantly different between the two sets of firms. Overall, these results are consistent with our prediction that performance suffers when firms make affiliated donations.

7. Conclusion

This paper shows that corporate donations to charities affiliated with independent directors reduce board monitoring effectiveness. In particular, we find higher excess CEO pay at firms making affiliated donations than at firms not making affiliated donations. The effect of affiliated donations on CEO compensation is much stronger when the compensation committee chair or a large fraction of compensation committee members are involved.

Our results are robust to tests alleviating endogeneity concerns: running change regressions, including various fixed effects, using propensity score matching, and using an instrumental variable specification. Our results also hold after we control for measures motivated by alternative hypotheses, such as free cash flow and director–CEO social ties. Our interpretation of the agency view is reinforced by additional findings that poorly performing CEOs are more likely to retain their jobs at firms that make large affiliated donations or donate to charities affiliated with more independent directors and that firm performance suffers as a result.

In summary, our paper uncovers a new factor that compromises a director's independence in addition to those currently recognized by the corporate governance literature (related-party transactions, social connections, and director–appointment decisions). We call for more disclosures regarding corporate donations affiliated with independent directors in SEC filings such as firms' proxy statements, regardless of their amount.

Appendix A. An Example of Forced Turnovers among College Head Coaches

Forced turnovers of college head coaches occur frequently. They often entail significant costs, including (1) an increase in the compensation package for the incoming coach (relative to the outgoing coach), (2) severance pay for the outgoing coach, (3) buyout for the incoming coach if there is a contractual obligation to the previous employer, (4) funds for firing/hiring assistant coaches associated with the turnover, and in some cases (5) associated legal expenses. The significant costs of firing a head coach increase the pressure to fundraise and the need for charitable donations, exogenous to firm characteristics that might affect CEO compensation.

In December 2017, Texas A&M (A&M) hired Jimbo Fisher away from Florida State University (FSU). A&M signed an incredible contract with Fisher and considered it a valuable investment:

²⁹ The economic impact of affiliated donations on Tobin's Q is similar to that of other governance measures in the prior literature. For example, in Doidge, Karolyi, and Stulz (2004), the coefficient for *Cross-list* ranges between 0.16 and 0.20 in Tobin's Q regressions, corresponding to 10.4% to 13% of the mean value of 1.54 for cross-listed firms. By comparison, the coefficient of 0.239 for affiliated donations corresponds to 11.8% of its mean value of 2.02.

1. A&M fired Kevin Sumlin without cause to hire Jimbo Fisher. A&M paid \$10.4 million (as a lump sum within sixty days of termination) to Sumlin to buy out the two years remaining in his contract, even though Sumlin became the head coach at the University of Arizona within two months.
2. Fisher had been the head coach of FSU since 2010 and had two more years remaining under his contract. A&M paid \$5 million to FSU to buy out Fisher's remaining commitment.
3. A&M granted Fisher a ten-year contract totaling \$75 million, fully guaranteed. If A&M fires Fisher without cause after a year, for example, it must pay Fisher \$67.5 million as severance.
4. Fisher can resign on his own without any financial obligation to A&M.
5. Fisher can receive annual bonuses of up to \$2.4 million based on performance.
6. Fisher is entitled to a variety of perks, including the use of two luxury vehicles, one country club membership, twelve to sixteen football and four basketball season tickets, and the use of private aircraft at a cost not to exceed \$300,000 annually.
7. Each year, Fisher is allocated a minimum salary pool of \$5 million for hiring assistant coaches. The salary pool will be adjusted annually to remain competitive.

It is worth noting that in 2017 A&M's athletic department's total revenue was \$212 million, \$93 million of which came from charitable contributions.

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Appendix B. Variable Definitions

Donation variables	
D(Affiliated donations)	Indicator that equals one if a firm donates to at least one charity affiliated with one or more of its independent directors in a given year and zero otherwise.
Ln(1+affiliated donations in thousands of dollars)	Logarithm of 1 plus all donations made to charities affiliated with a firm's independent directors in a given year, in thousands of dollars.
Ln(0.001+affiliated donation amount, scaled by average director pay)	Logarithm of 0.001 plus the ratio of affiliated donation amount to average director pay.
Ln(0.001+affiliated donation amount, scaled by total annual donation of the firm)	Logarithm of 0.001 plus the ratio of affiliated donation amount to the amount of total charitable donations made by the firm that year.
D(Affiliated donation related to compensation committee)	Indicator that equals one if a firm donates to at least one charity affiliated with at least one independent director who serves on the firm's compensation committee and zero otherwise.
D(Affiliated donations related to compensation committee chair)	Indicator that equals one if a firm donates to at least one charity affiliated with the compensation committee chair and zero otherwise.
D(Affiliated donations related to other compensation committee member)	Indicator that equals one if a firm donates to at least one charity affiliated with at least one non-chair member of the compensation committee and zero otherwise.
D(Affiliated donations unrelated to compensation committee)	Indicator that equals one if a firm donates to at least one charity affiliated with at least one independent director who does not serve on the compensation committee in a given year and zero otherwise.
D(Initiation of affiliated donations)	Indicator that equals one if a firm donates to at least one charity affiliated with at least one independent director for the first time and zero otherwise.

D(Initiation of affiliated donations related to compensation committee)	Indicator that equals one if a firm makes its first affiliated donation that year and the donation goes to a charity affiliated with an independent director on the compensation committee and zero otherwise.
D(Initiation of affiliated donations unrelated to compensation committee)	Indicator that equals one if a firm makes its first affiliated donation that year and the donation goes to a charity affiliated with an independent director outside the compensation committee and zero otherwise.
D(Termination of affiliated donations)	Indicator that equals one if a firm stops donating to all charities that are affiliated with any independent directors at once and zero otherwise.
D(Termination of affiliated donations related to compensation committee)	Indicator that equals one if a firm stops all affiliated donations and the last donation goes to a charity affiliated with independent director(s) on the compensation committee and zero otherwise.
D(Termination of affiliated donations unrelated to compensation committee)	Indicator that equals one if a firm stops all affiliated donations and the last donation goes to a charity affiliated with independent director(s) outside the compensation committee and zero otherwise.
D(Unaffiliated donations)	Indicator that equals one if a firm donates only to charities not affiliated with any of its independent directors and zero otherwise.
D(CEO-affiliated donations)	Indicator that equals one if a firm makes donation(s) to at least one charity affiliated with the CEO and zero otherwise.
D(Inside director-affiliated donations)	Indicator that equals one if a firm makes donation(s) to at least one charity affiliated with inside director(s) and zero otherwise.

CEO compensation, instrumental variable, and forced CEO turnover

Total pay	Total direct annual compensation (ExecuComp variable <i>TDC1</i>) in thousands of dollars. <i>TDC1</i> is calculated as <i>Salary+Bonus+Other Annual+LTIP Payouts+Restricted Stock Grants+Value of Options Granted+All Other</i> prior to December 2006, and <i>Salary+Bonus+Nonequity Incentive Plan Compensation+Grant Date Fair Value of Stock Awards+Value of Options Granted+Other Compensation</i> starting in December 2006.
D(Forced coach turnovers)	Indicator that equals one if multiple independent directors' affiliated universities force out the head coach of their basketball or football team in a given year and zero otherwise.
D(Forced CEO turnovers)	Indicator that equals one if the CEO is fired in a given year and zero otherwise. Whether a CEO is forced out is determined based on Parrino (1997) and Jenter and Kanaan (2015). For more details about the turnover data, see Gao et al. (2017).

Corporate governance

D(CEO serves as chairman)	Indicator that equals one if the CEO is the Chairman of the Board and zero otherwise.
CEO tenure	No. of years elapsed since the CEO assumed his or her current position.
CEO ownership	CEO's stock ownership in percentage terms.
Board size	The total number of members on the board.
Board independence	The percentage of independent board members. Independent directors are directors that are not affiliated with the company according to the RiskMetrics (formerly IRRC) definition.
D(Independent board)	Indicator that equals one if there are fewer than three nonindependent directors on the board and zero otherwise.
Busy director	A director who serves on three or more corporate boards, based on information provided in BoardEx.
D(Social connection(s) between independent directors and CEO)	Indicator that equals one if any independent director is connected to the CEO through prior work (for-profit or nonprofit) or education and zero otherwise.
D(Above-median fraction of co-opted directors)	Indicator that equals one if the fraction of co-opted directors is above the median and zero otherwise. A director is co-opted if she is hired after the current CEO assumes his or her position.
Top five institutions' ownership	Total ownership by the five institutions with the largest holdings of the firm's stock.
D(CEO is connected to independent director(s) via affiliated charity)	Indicator that equals one if any independent director is connected to the CEO through charities receiving affiliated donations and zero otherwise.

Firm financials

Ln(Assets)	Logarithm of total book assets (measured in millions of dollars).
Stock return	Annual stock return as reported in ExecuComp.
Profit margin	Operating income before depreciation, divided by sales.

ROA	Operating income before depreciation, divided by total book assets.
dROA	Change in ROA from the previous year.
Market-to-book ratio (Tobin's Q)	The sum of the market value of equity and total book assets minus total common equity, all divided by total book assets. The market value of equity is the fiscal-year-end stock price multiplied by total number of shares outstanding.
Stock return volatility	The standard deviation of daily stock returns in each year.
Debt/Assets	The sum of long-term debt and debt in current liabilities divided by total book assets.
Advertisement/sales	Advertisement expenditure divided by sales.
R&D/assets	Research and development expenditure divided by book value of assets.
Sales growth	Change in the logarithm of sales from the previous year.
Industry growth	Two-digit SIC industry sales growth rate.
State corporate tax rate	State top statutory corporate tax rate.
Free cash flow	Operating income before depreciation minus interest expenses and income taxes and capital expenditures, scaled by total assets, following Masulis et al. (2007).
Ln(Sales)	Logarithm of sales (measured in millions of dollars).
Ln(Market equity)	Logarithm of market equity (measured in millions of dollars).
Industry cash flow volatility	Industry-year median cash flow volatility. Cash flow volatility is the standard deviation of quarterly operating cash flows in the previous six fiscal years.

Director characteristics

Ln(No. of charities director is affiliated with)	Logarithm of the number of charities the director is affiliated with.
Ln(No. of S&P 500 boards director sits on)	Logarithm of the number of S&P 500 firms' boards the director sits on.
D(Director is an NEO at another S&P 500 firm)	Indicator that equals one if the director is a named executive officer at another S&P 500 firm and zero otherwise.
D(Social connections between director and CEO)	Indicator that equals one if the director and CEO have any social connections and zero otherwise.
D(Director is independent chairman)	Indicator that equals one if the director is the independent chairman of the firm and zero otherwise.
D(Director is lead director)	Indicator that equals one if the director is a lead director of the firm and zero otherwise.
D(Director is chair of a board committee)	Indicator that equals one if the director is the chair of a committee on the firm's board and zero otherwise.
Ln(1+director pay in \$1,000)	Logarithm of one plus the director's total pay in thousands of dollars. Available since 2006.
D(Director pay is in the top decile)	Indicator that equals one if the director's total pay is in the top decile of the sample and zero otherwise.
D(Director has prior industry experience)	Indicator that equals one if the director has prior experience (either as a director or as a senior manager) in the same industry as the firm and zero otherwise.
Director age	The director's age.
Director tenure in excess of CEO tenure	No. of years by which the director's tenure exceeds the CEO's tenure in their respective current positions.
D(Director is female)	Indicator that equals one if the director is female and zero otherwise.

All variables (except indicators) are winsorized at the first and ninety-ninth percentiles.

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